



Population Range Estimation

Continuous-Time Animal Movement Modeling

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Overview

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- 1) Introduction
- 2) Kernel density estimation
- 3) Validation
- 4) Discussion

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Introduction

- In conservation, we are often concerned with **populations** rather than individuals

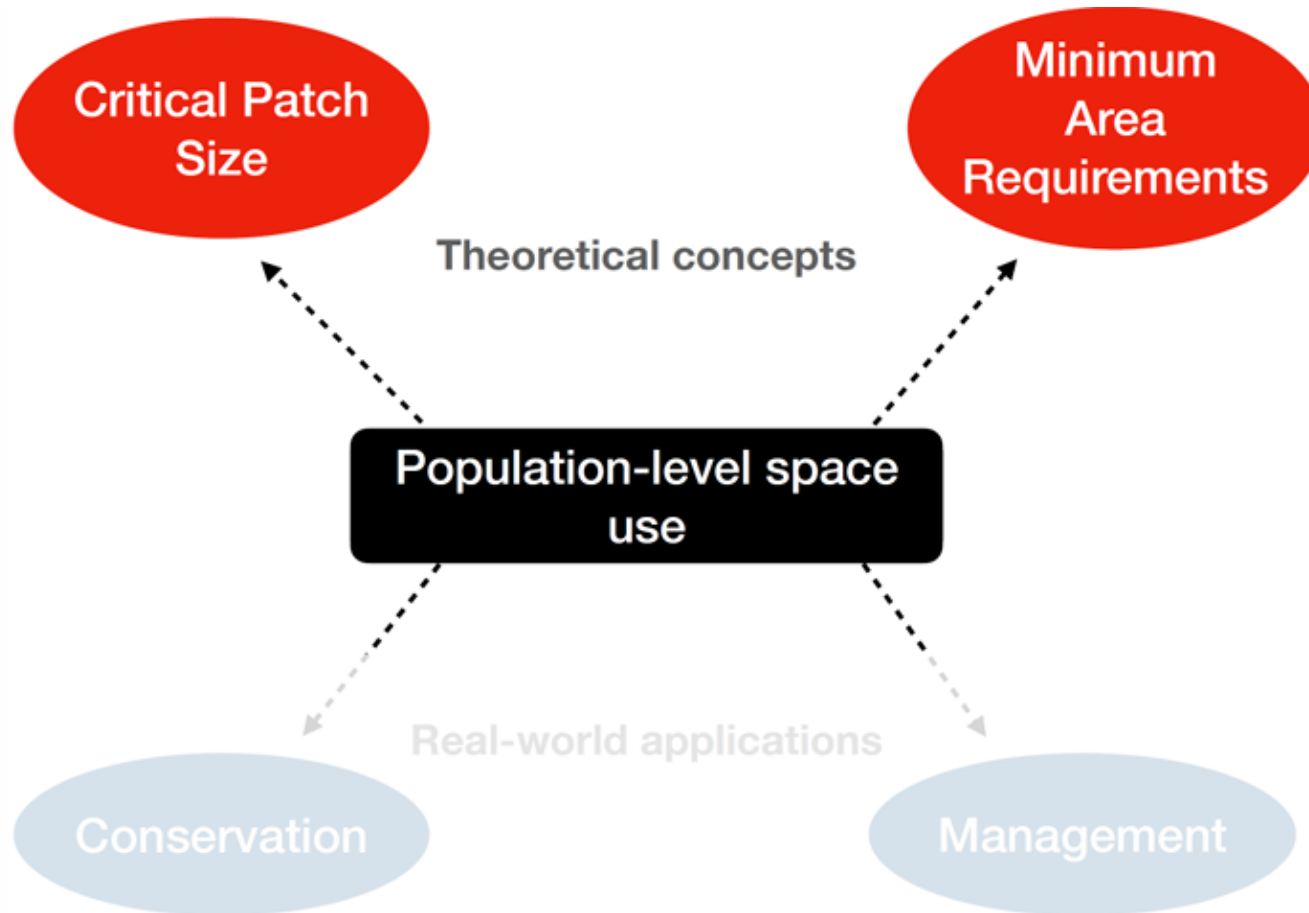


Simon Tonge

Eva Blue

How can we get population range estimates from individually tracked animals?

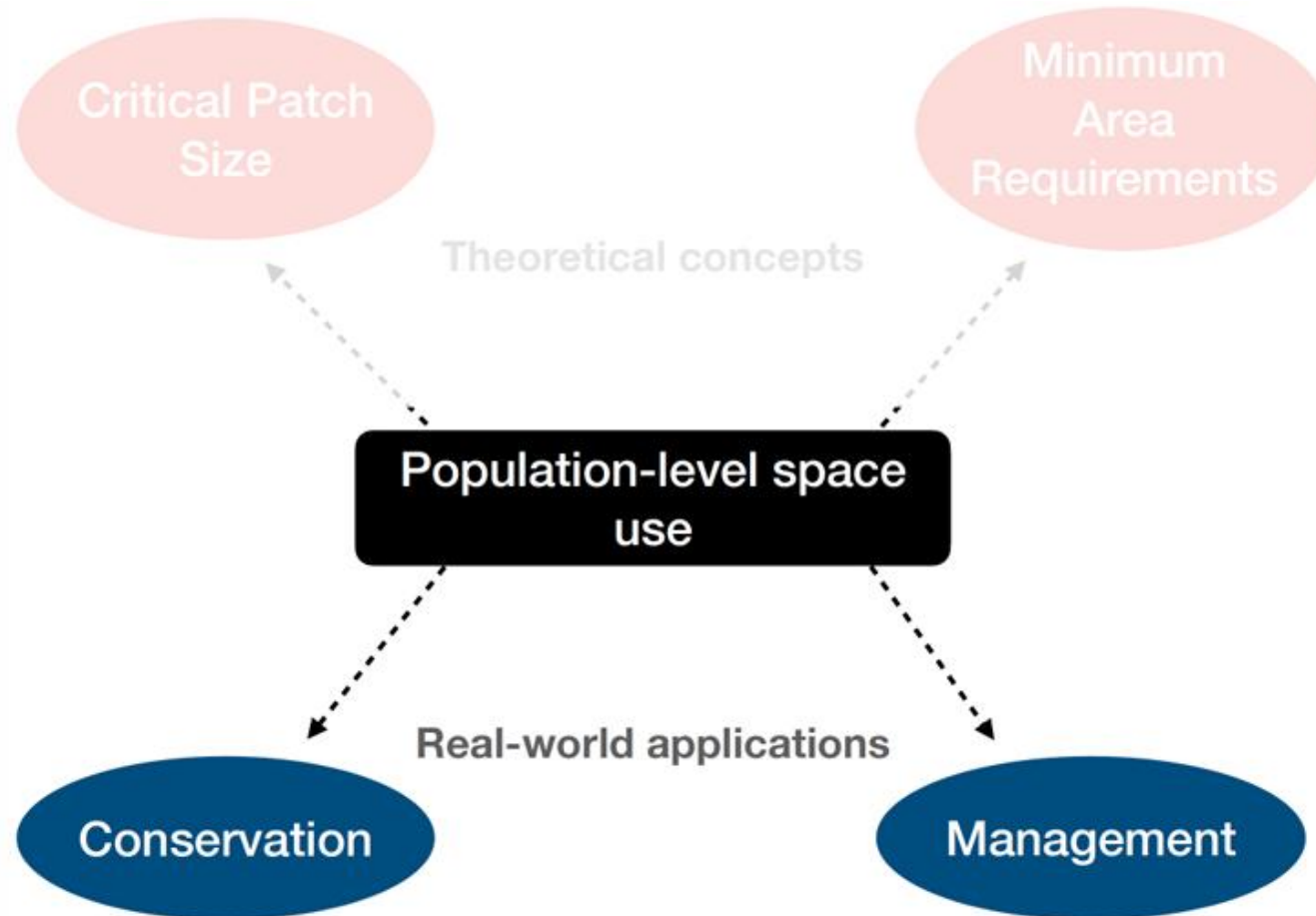
Applications of population range estimation



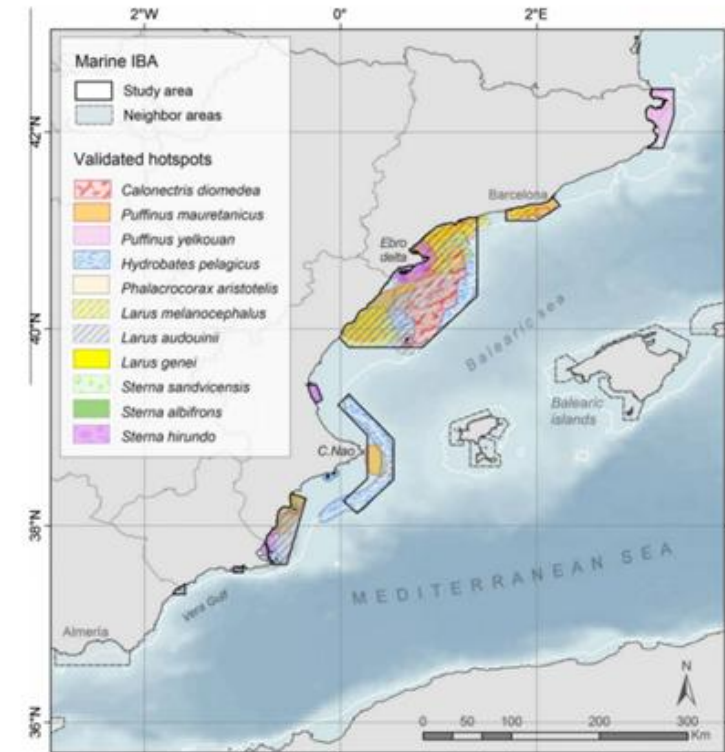
Example Ranges of Minimum Patch Area		
Taxa		Patch Area
 Plants		5 to ≥ 250 ac
 Invertebrates		50 sq ft to ≥ 2.5 ac
 Reptiles and Amphibians		3 to ≥ 35 ac
 Grassland Birds		12 to ≥ 135 ac
 Waterfowl		≥ 12 ac
 Forest Birds		5 to ≥ 95 ac
 Small Mammals		2.5 to ≥ 25 ac
 Large Mammals		40 ac to ≥ 2 sq mi
 Large Predator Mammals		3.5 to ≥ 850 sq mi

Source: USDA

Applications of population range estimation



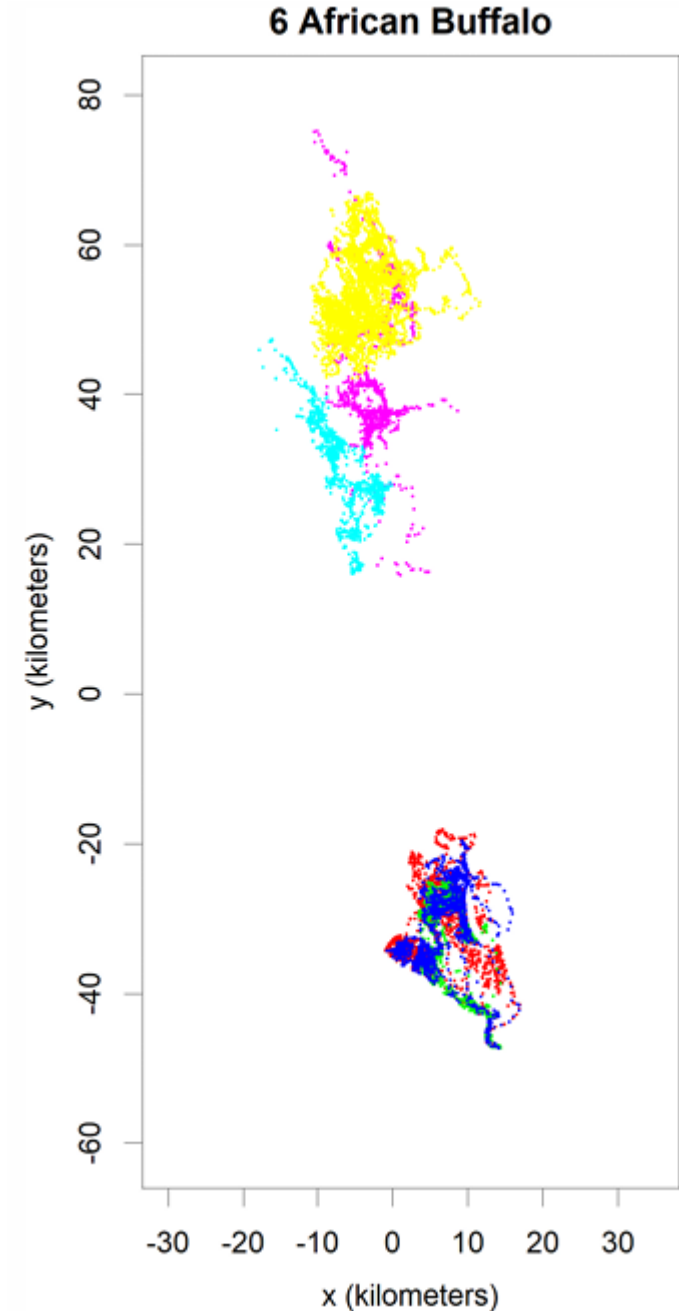
Seabird foraging hotspots used to define protected areas in Spain



Arcos et al. (2021)

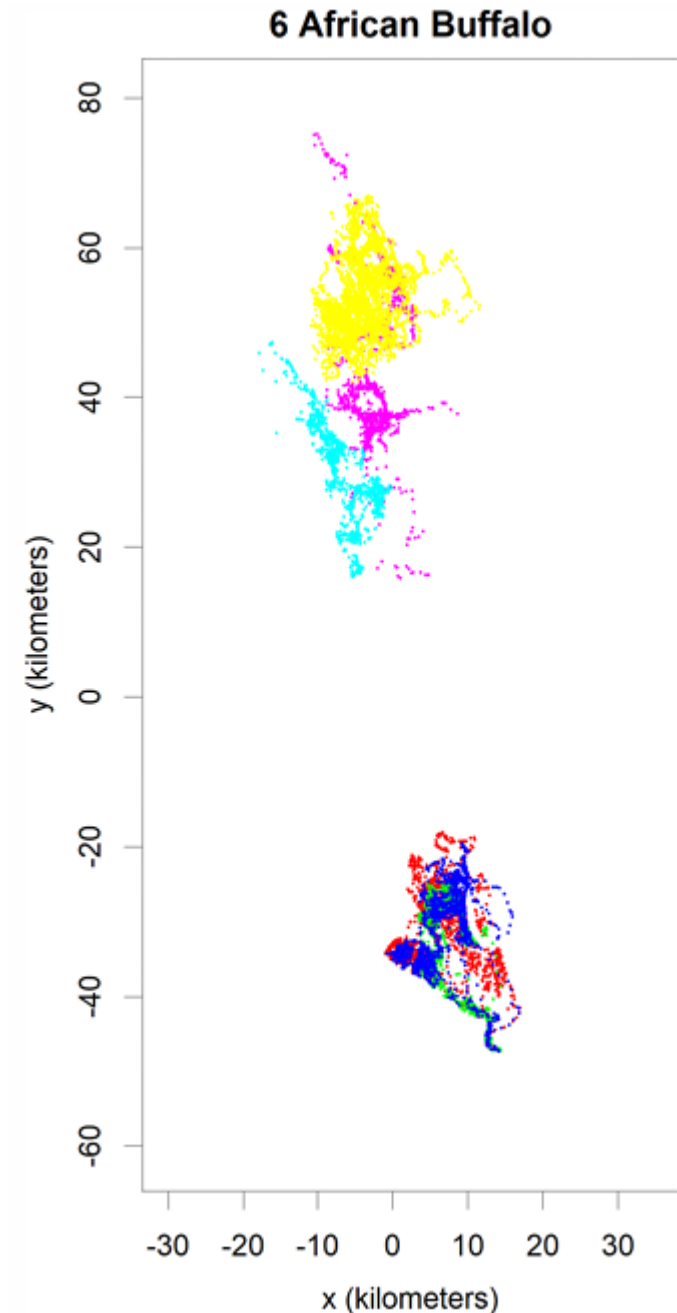
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- Tracks sampled from populations:
 - What's the population distribution?



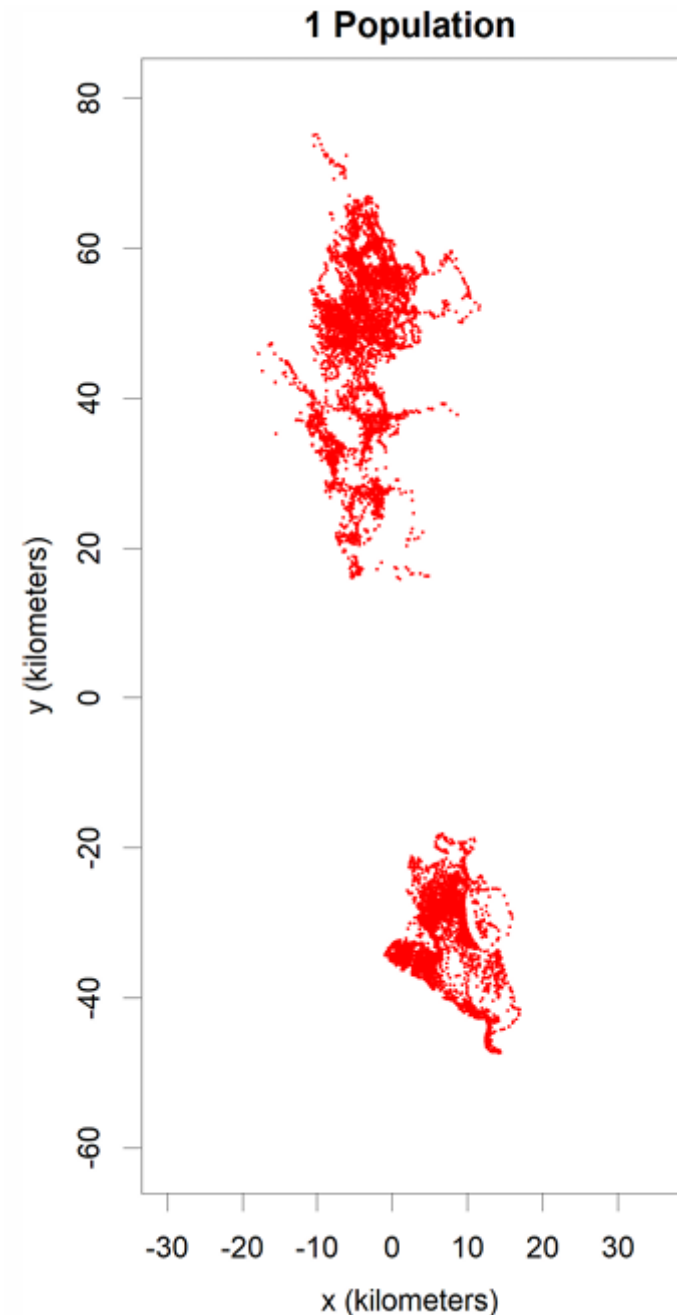
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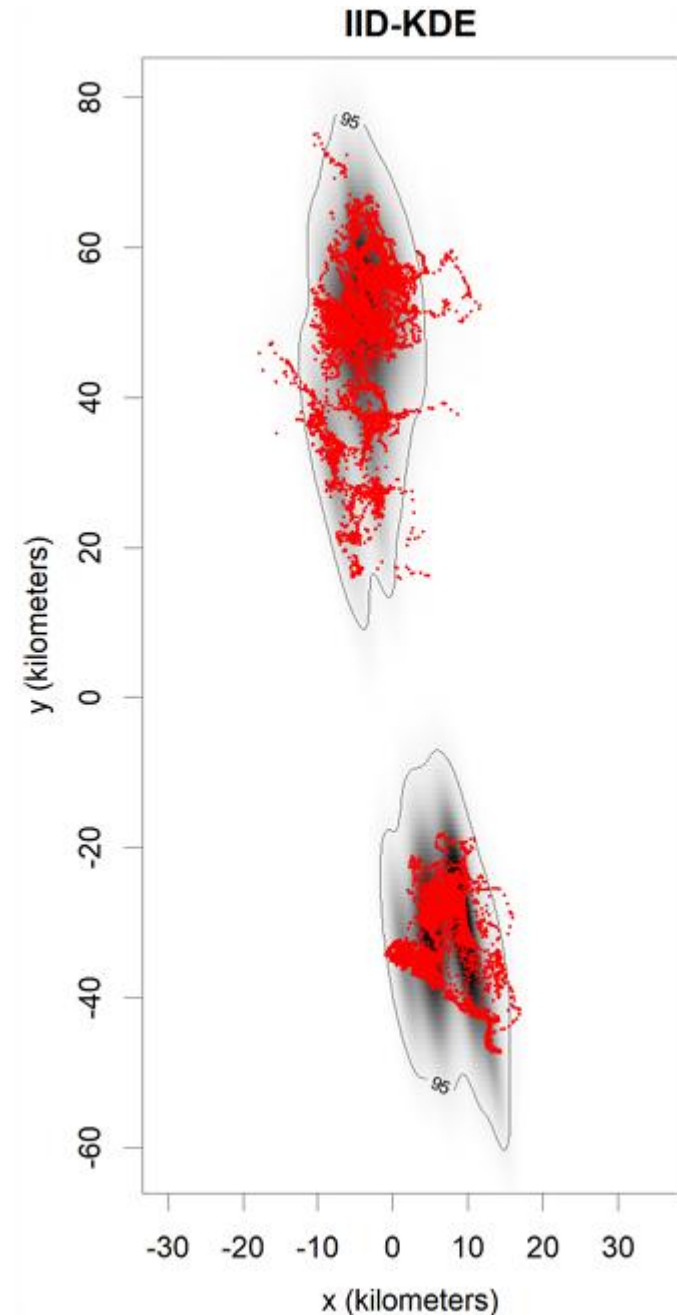
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- Methods:
 - KDE of population



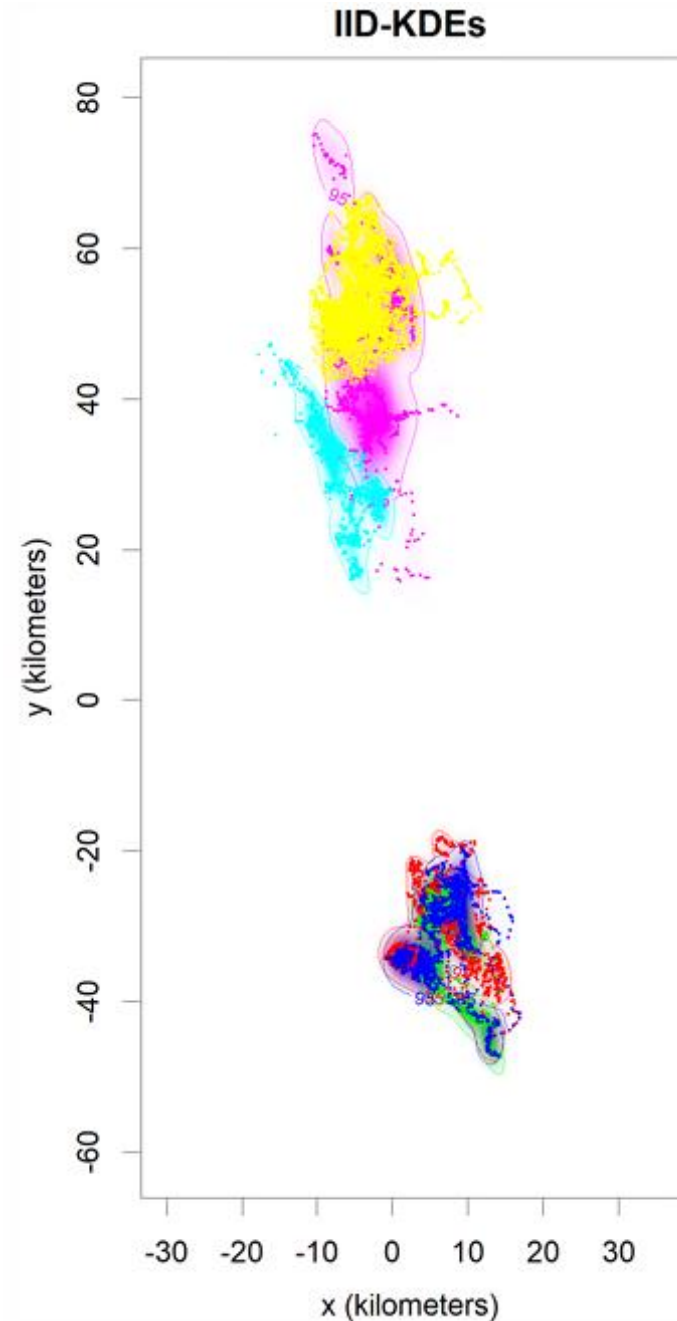
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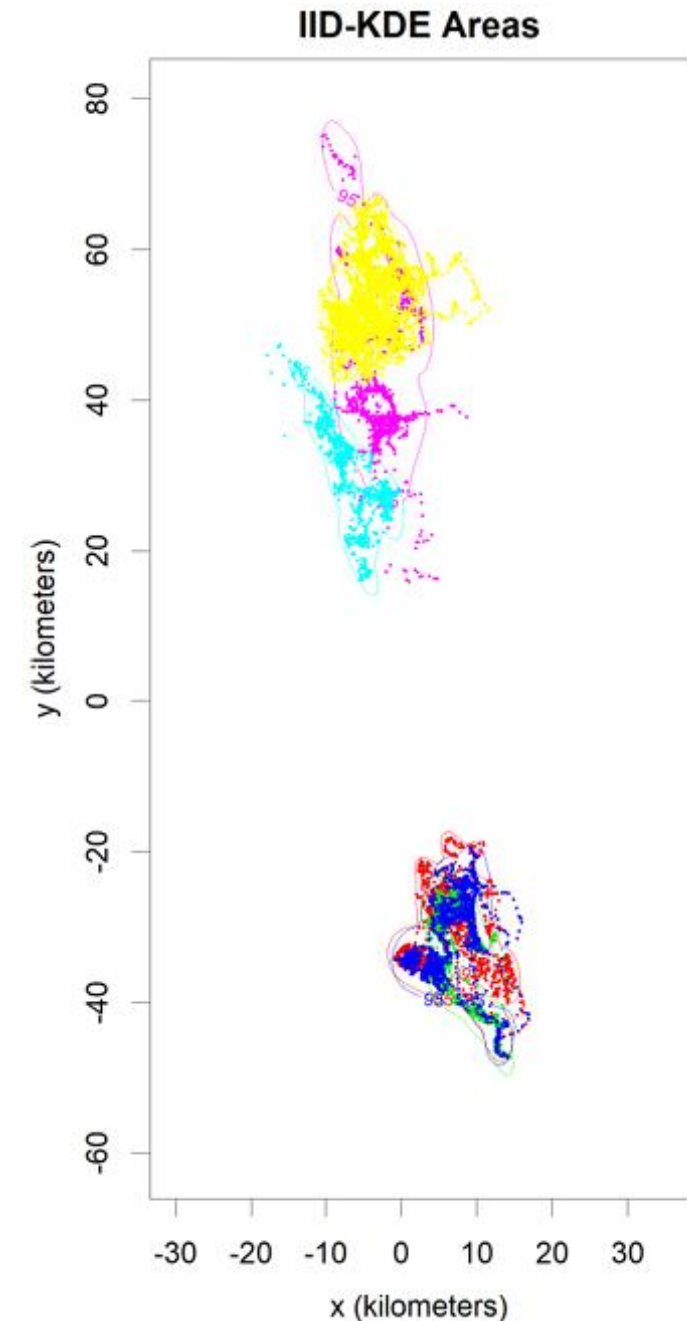
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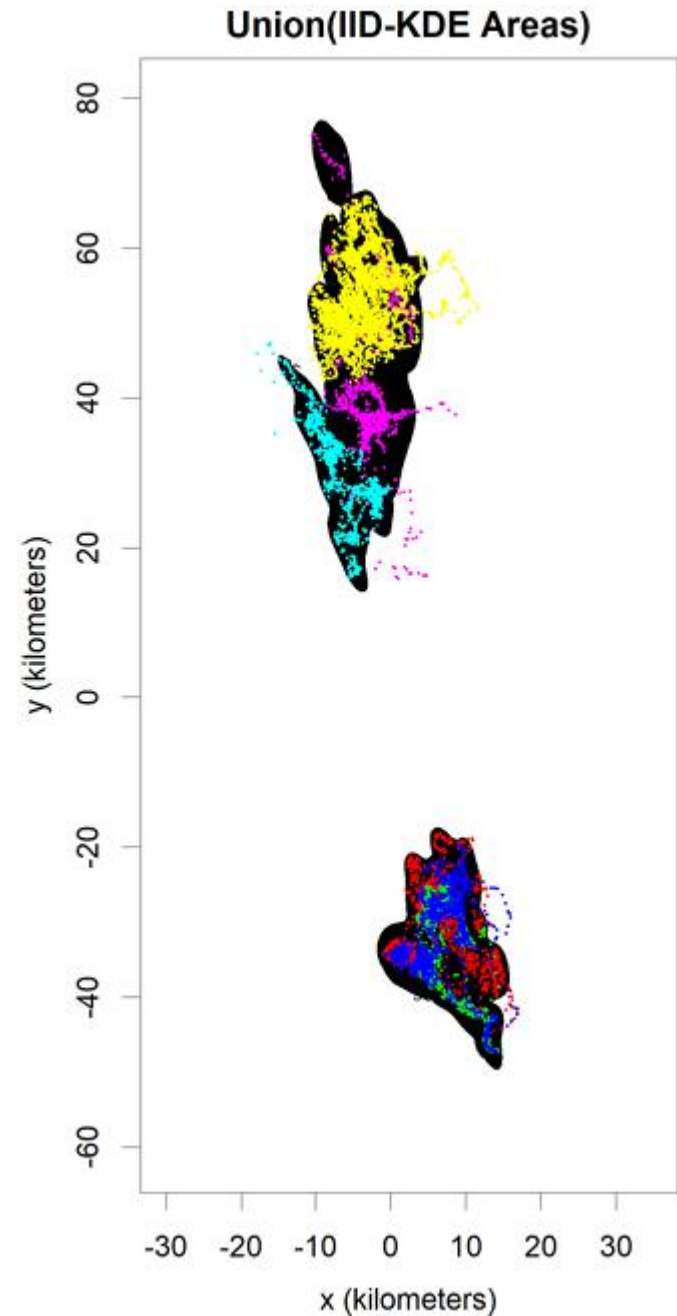
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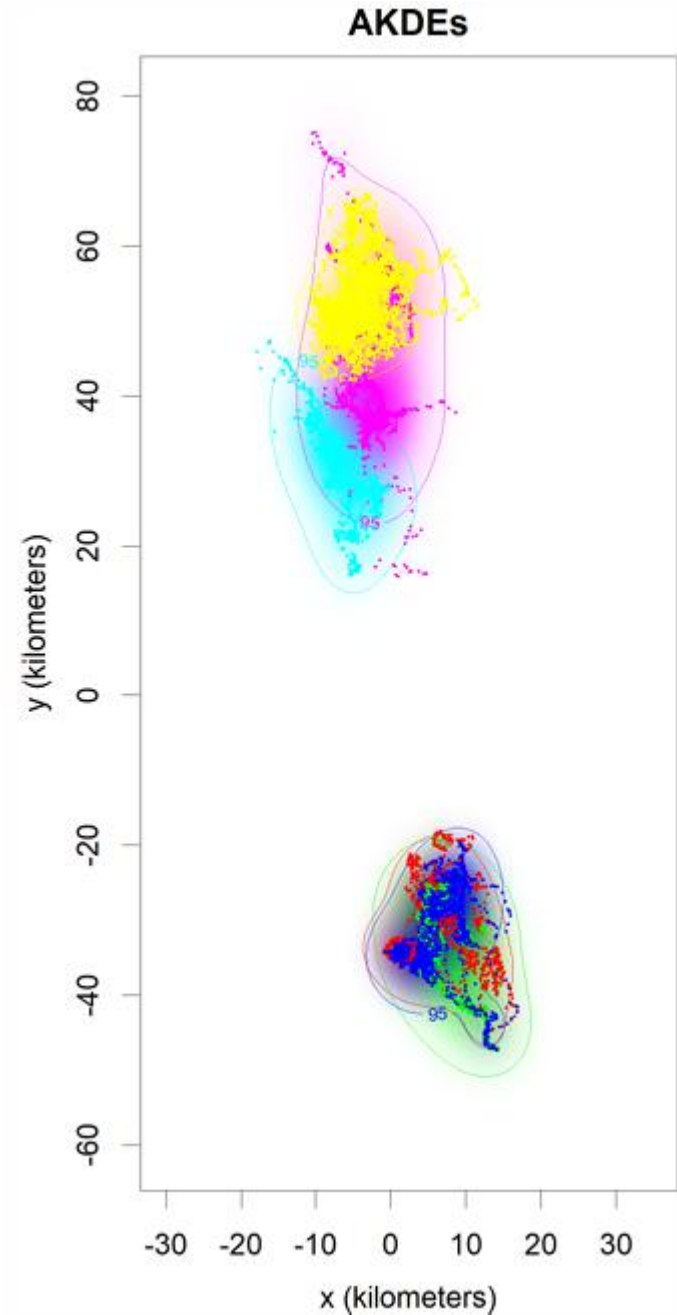
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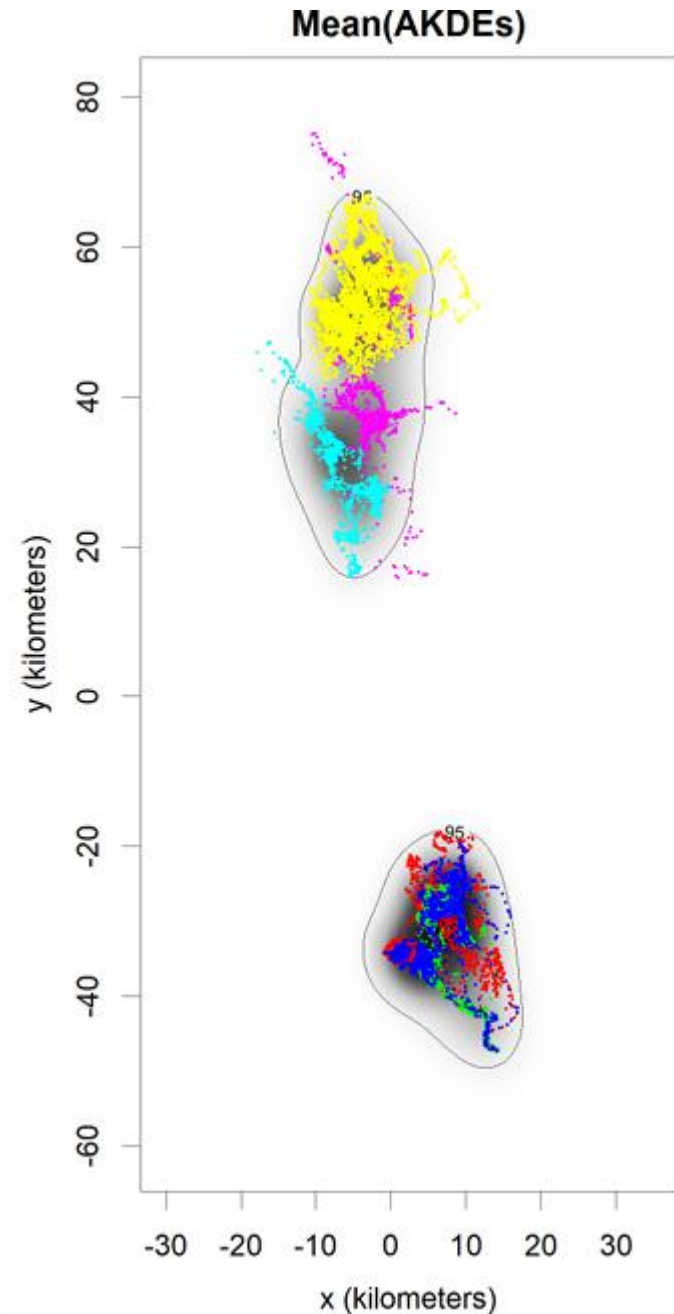
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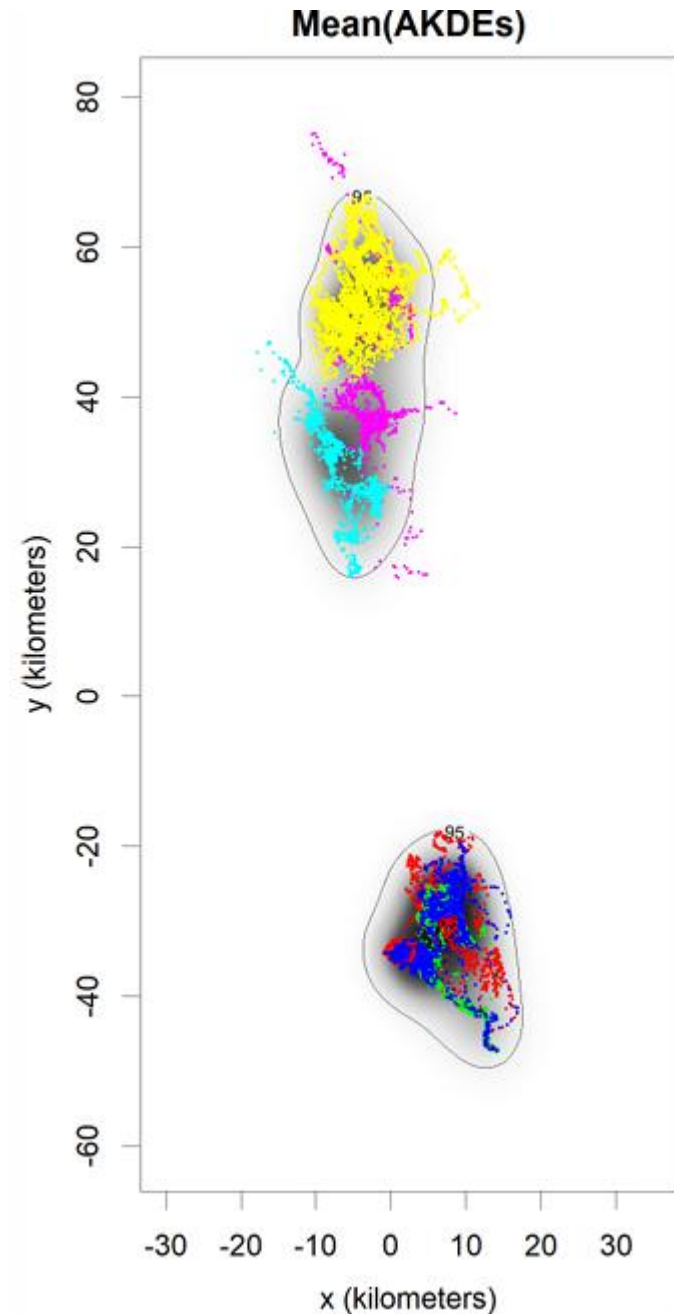
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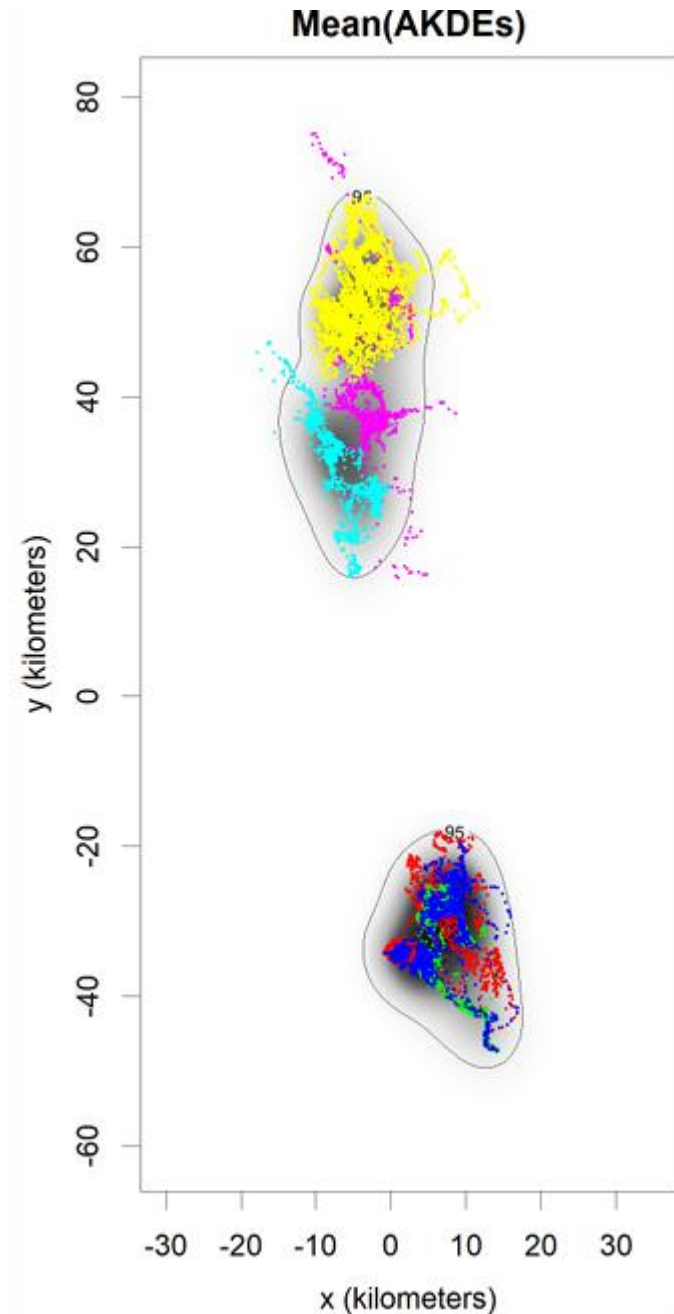
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- Dual challenge (unsolved)
 - Individual temporal autocorrelation
 - Population variation



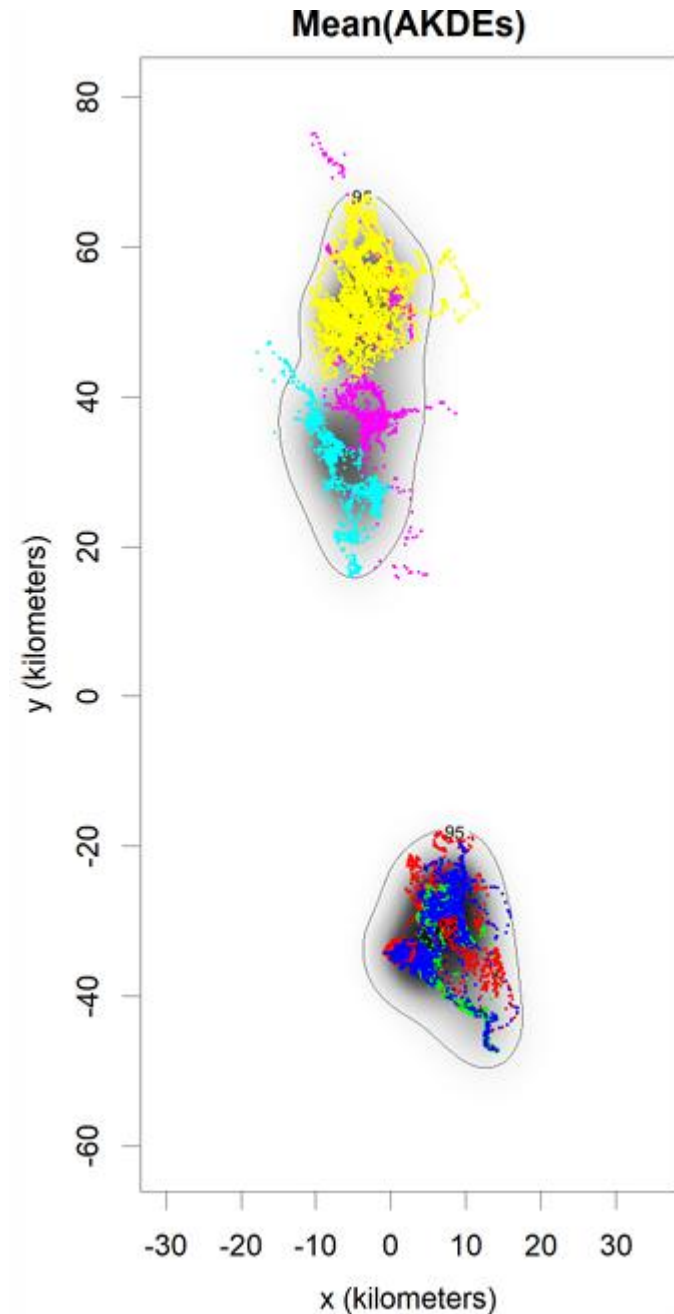
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- Sample size vs. saturation

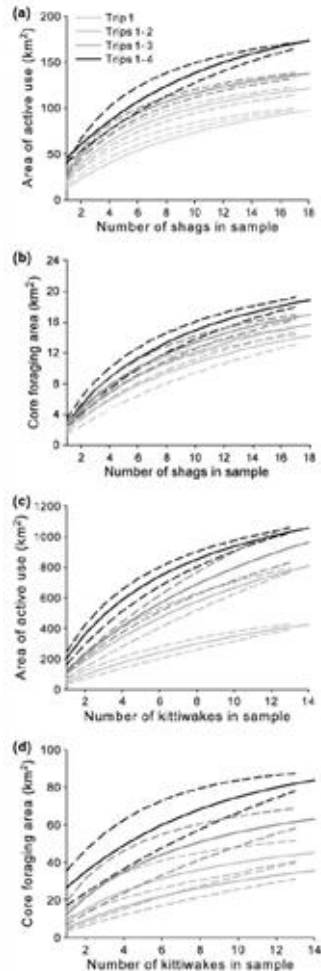


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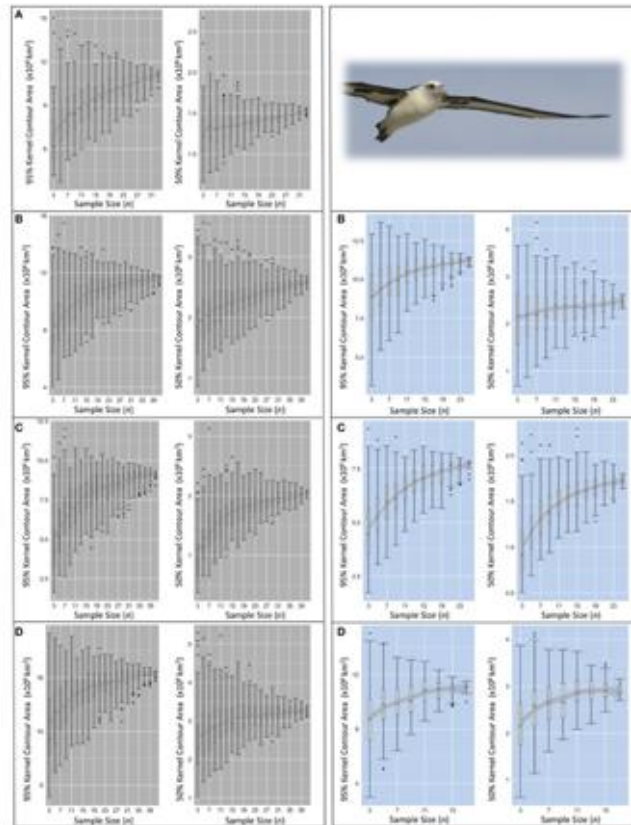
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- Sample size vs. saturation
 - Extrapolation of asymptote



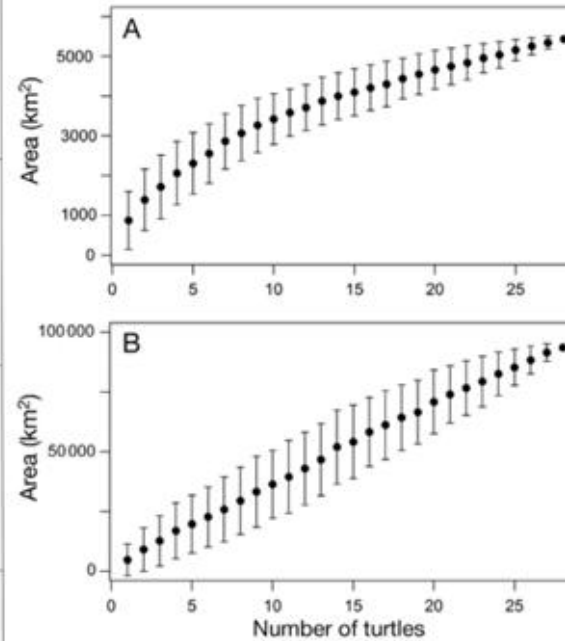
Sample size vs. Saturation



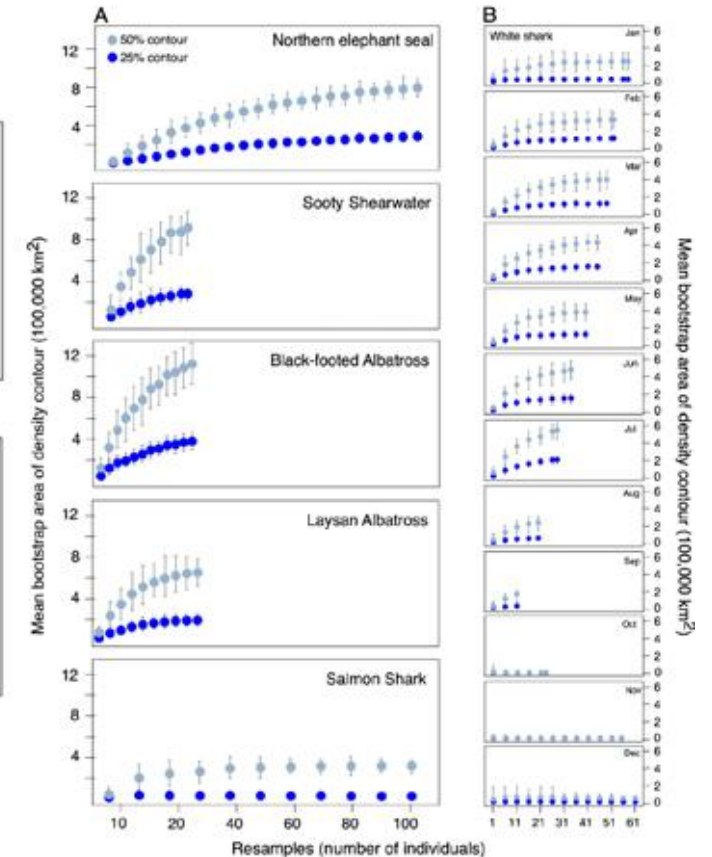
Soanes et al. (2013)



Gutowsky et al. (2015)

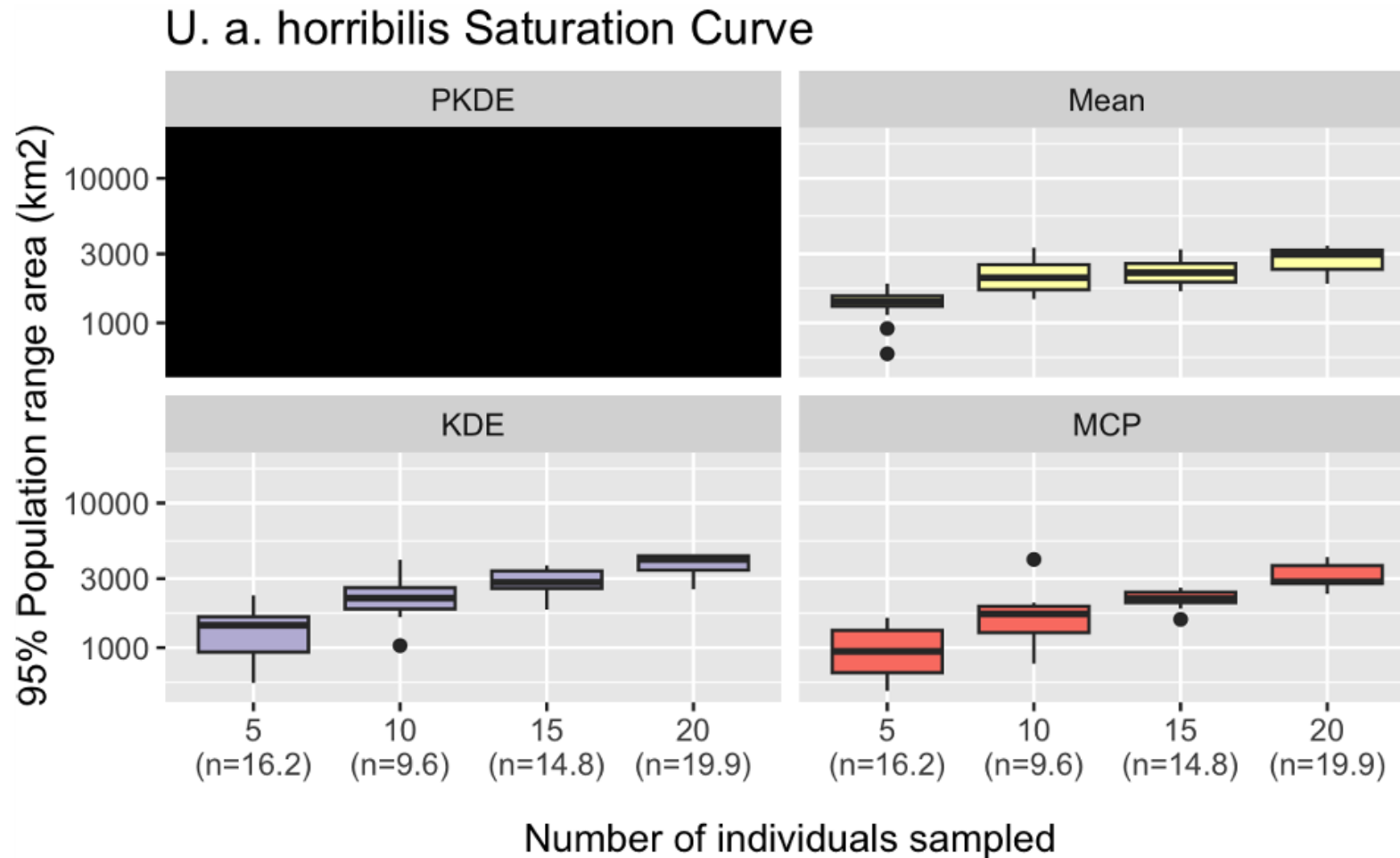


Thums et al. (2018)



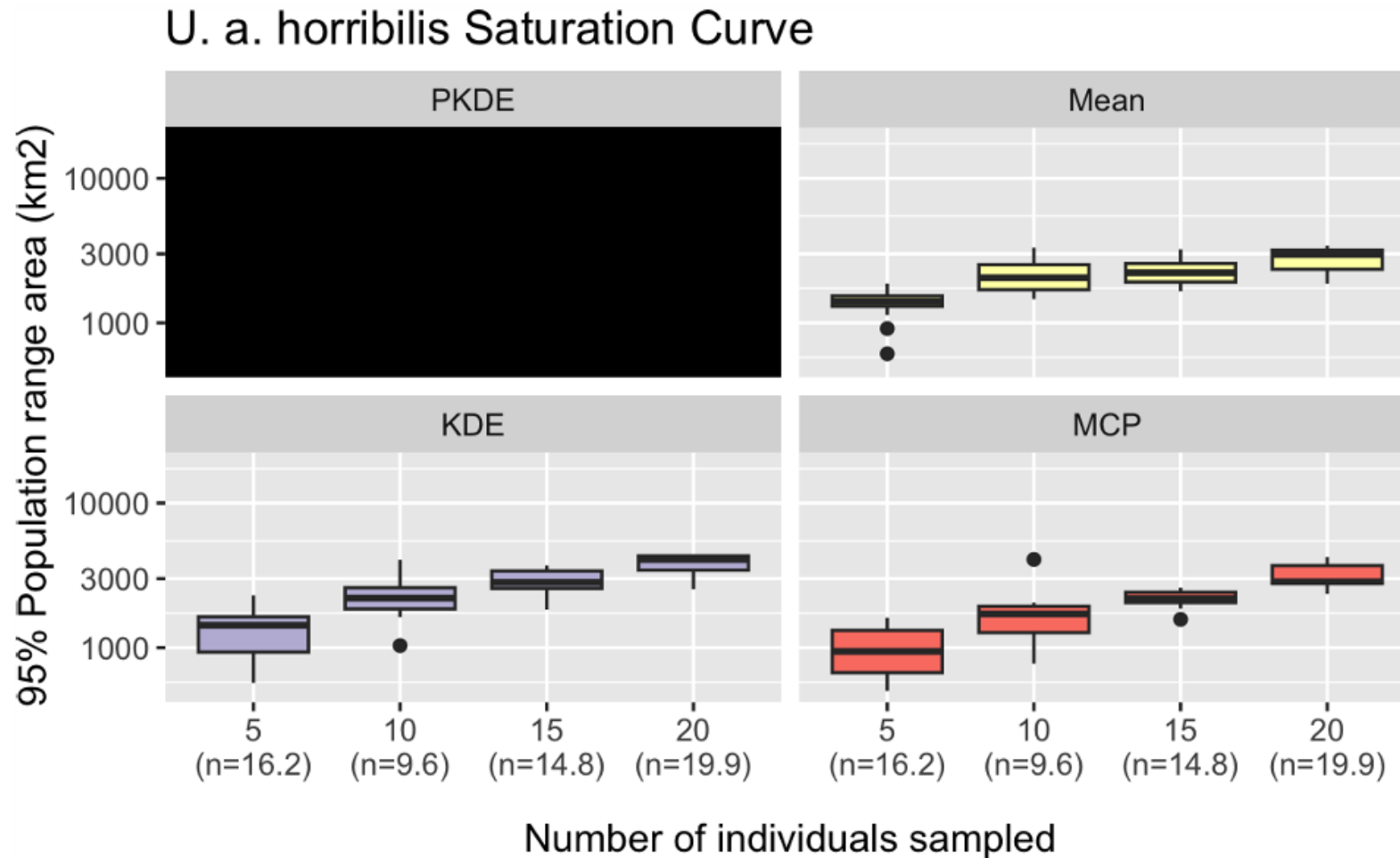
Sequeira et al. (2019)

Sample size vs. Saturation



Grizzly Bear
(*Ursus arctos horribilis*)

Sample size vs. Saturation



This bias is from NOT modeling population variance



Grizzly Bear
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Kernel Density Estimation

A KDE is a weighted average of kernels:

$$\hat{p}(\mathbf{x}|\mathbf{H}) = \sum_t^n w(t) \kappa(\mathbf{x} - \mathbf{x}(t)|\mathbf{H})$$

$\hat{p}(\cdot)$: probability density estimate

$\mathbf{x}$: location of interest

$w(t)$: weight at time t

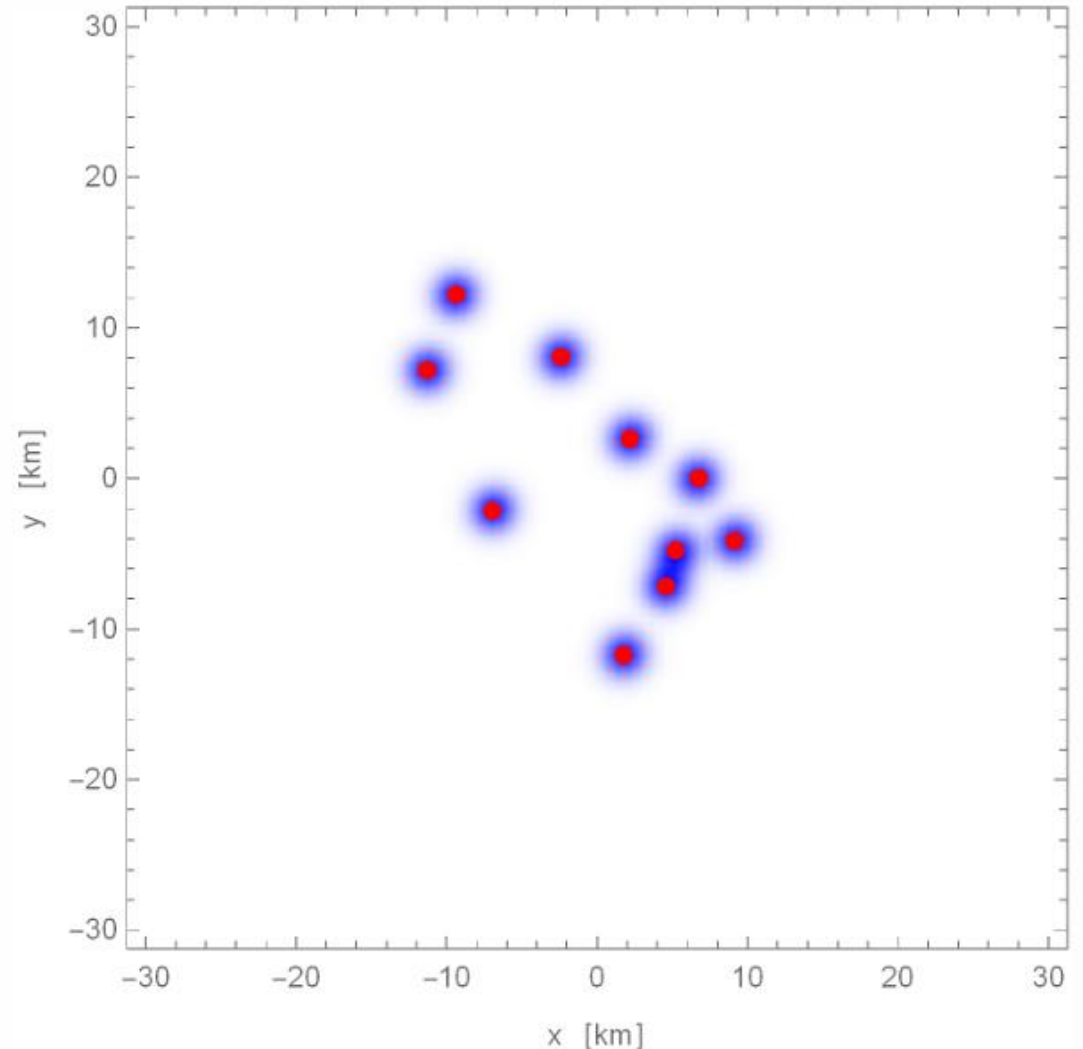
$\kappa(\cdot)$: kernel function (e.g., Gaussian)

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$\mathbf{H}$: bandwidth (e.g., COV of kernel, κ)

Where the optimal $\mathbf{H}$ minimizes the MISE:

$$\text{MISE}[\mathbf{H}] = \mathbb{E} \left[\iint (\hat{p}(\mathbf{x}|\mathbf{H}) - p(\mathbf{x}))^2 d\mathbf{x} \right]$$



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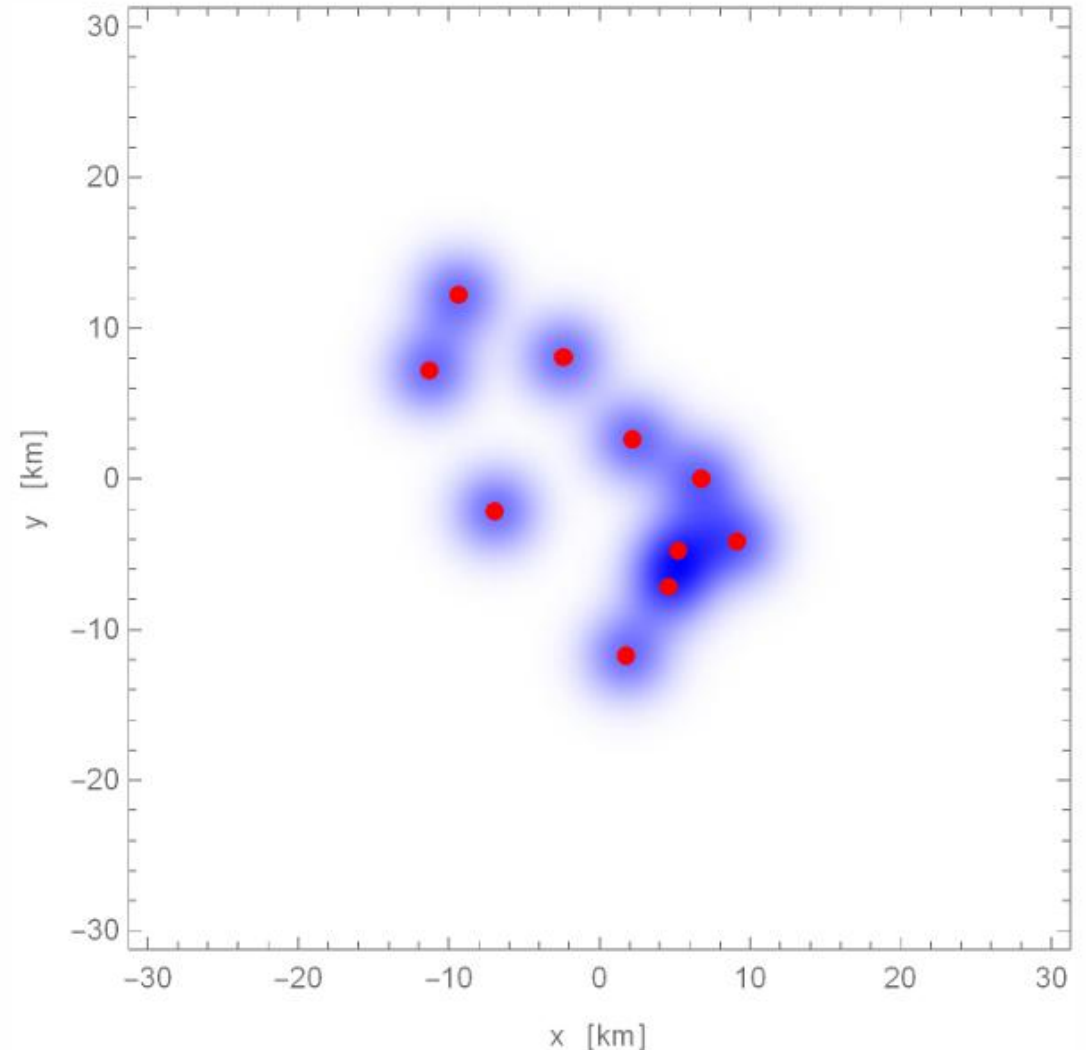
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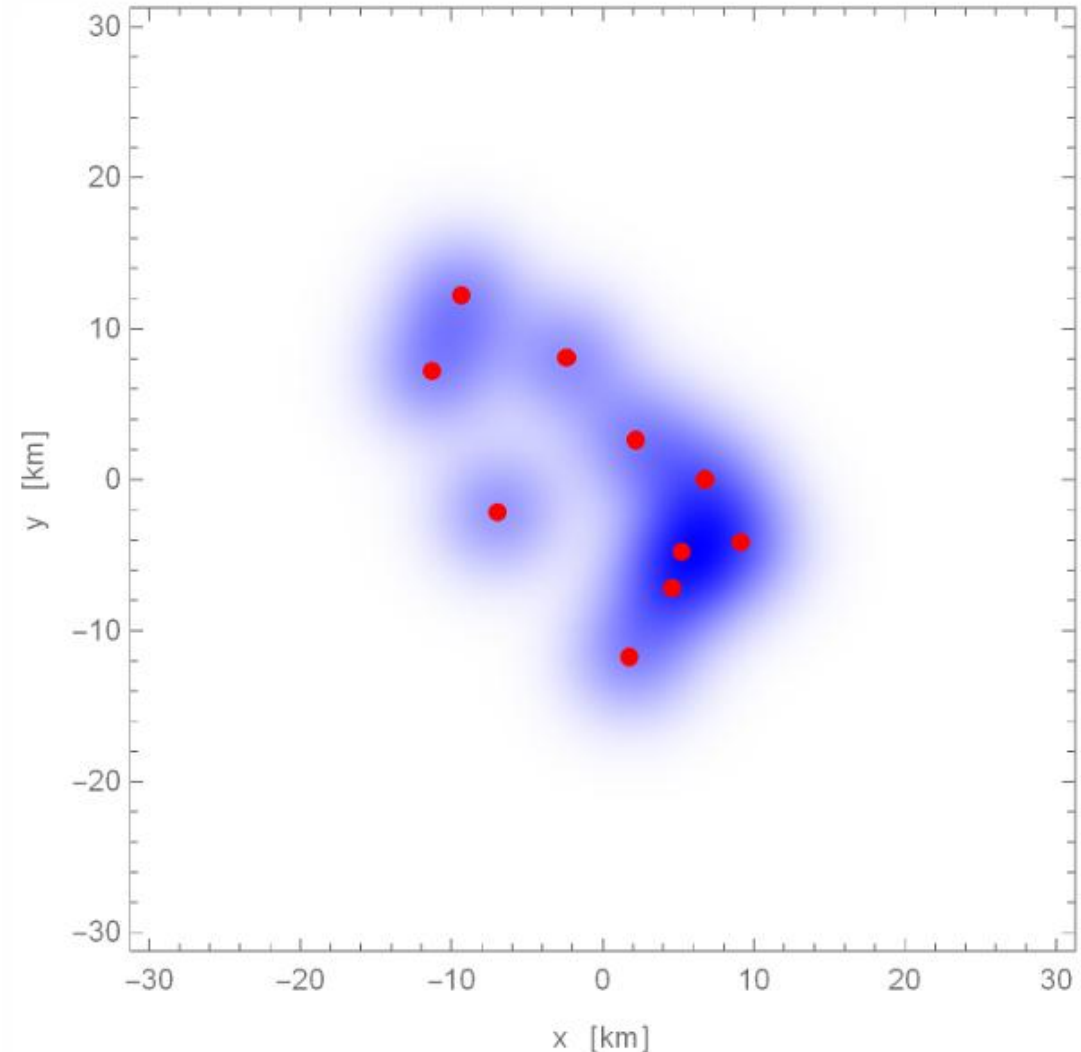
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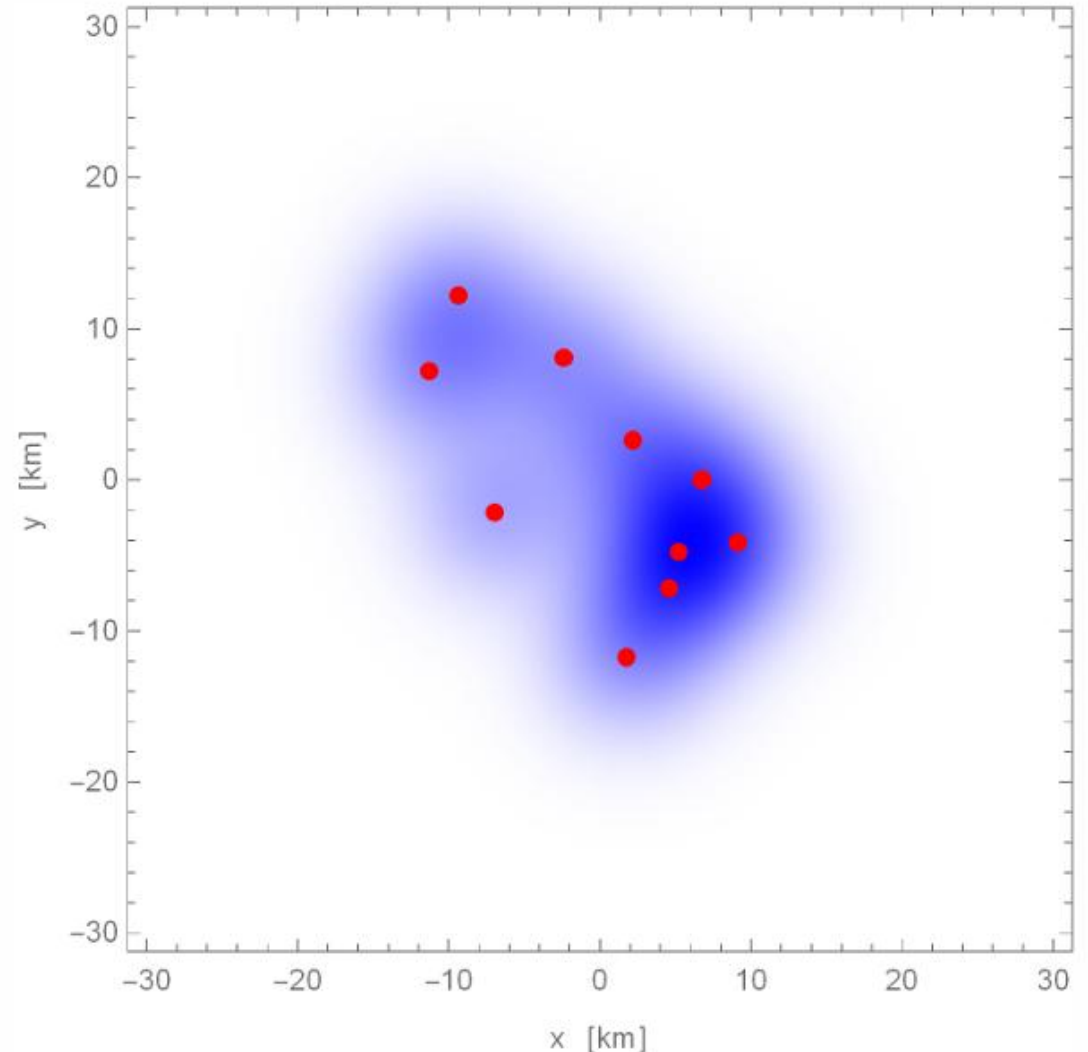
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Autocorrelated Kernel Density Estimation

$$\text{MISE}[\mathbf{H}] = E \left[\iint (\hat{p}(\mathbf{x}|\mathbf{H}) - p(\mathbf{x}))^2 d\mathbf{x} \right]$$

$p(\mathbf{x})$ = approximation (e.g., Gaussian reference function)

$\mathbf{x}(t) \sim \text{IID}$ (conventional KDE)

$\mathbf{x}(t) \sim \text{ACF}$ (autocorrelated KDE)

- Bandwidth optimization (Turlach, 1993)
- AKDE (Fleming et al., 2015)
- Bias-corrected AKDEc (Fleming & Calabrese, 2017)
- (MISE) optimally weighted wAKDE (Fleming et al., 2018)

Autocorrelated Kernel Density Estimation

$$\text{MISE}[\mathbf{H}] = \mathbb{E} \left[\iint (\hat{p}_{\text{pop.}}(\mathbf{x}|\mathbf{H}) - p_{\text{pop.}}(\mathbf{x}))^2 d\mathbf{x} \right]$$

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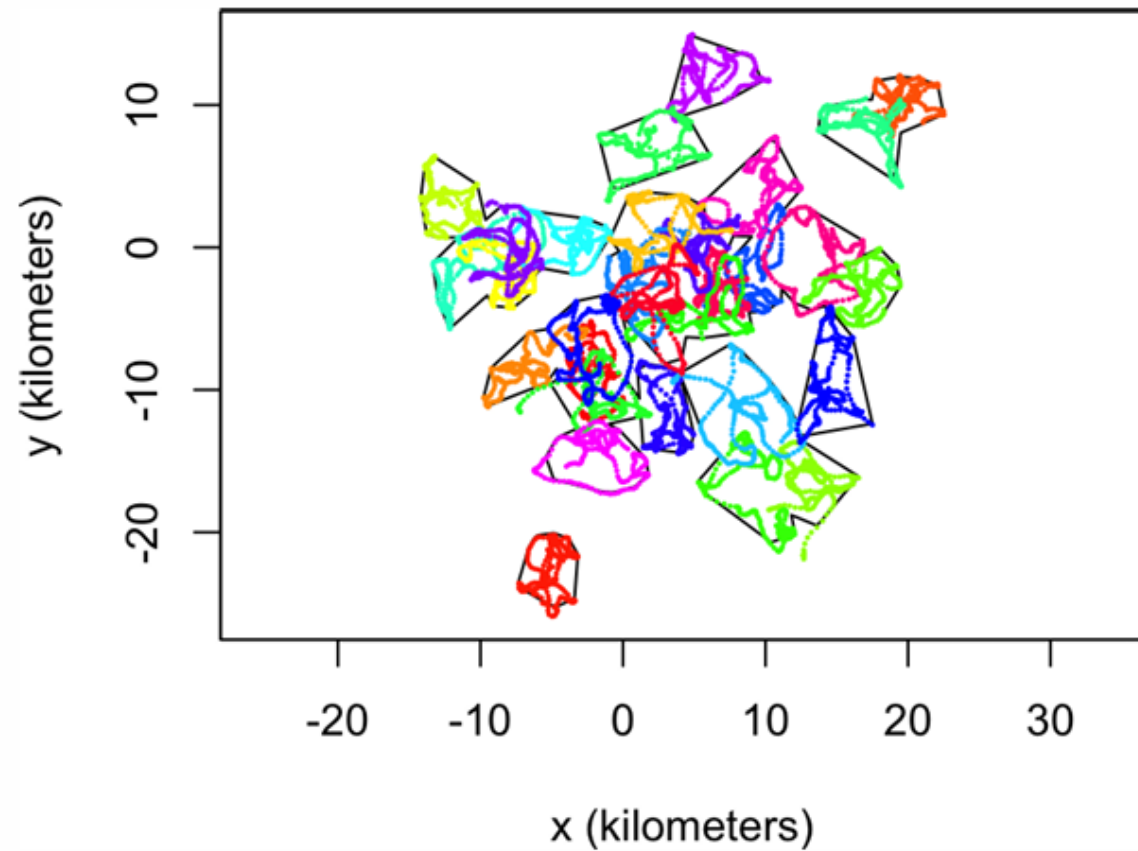
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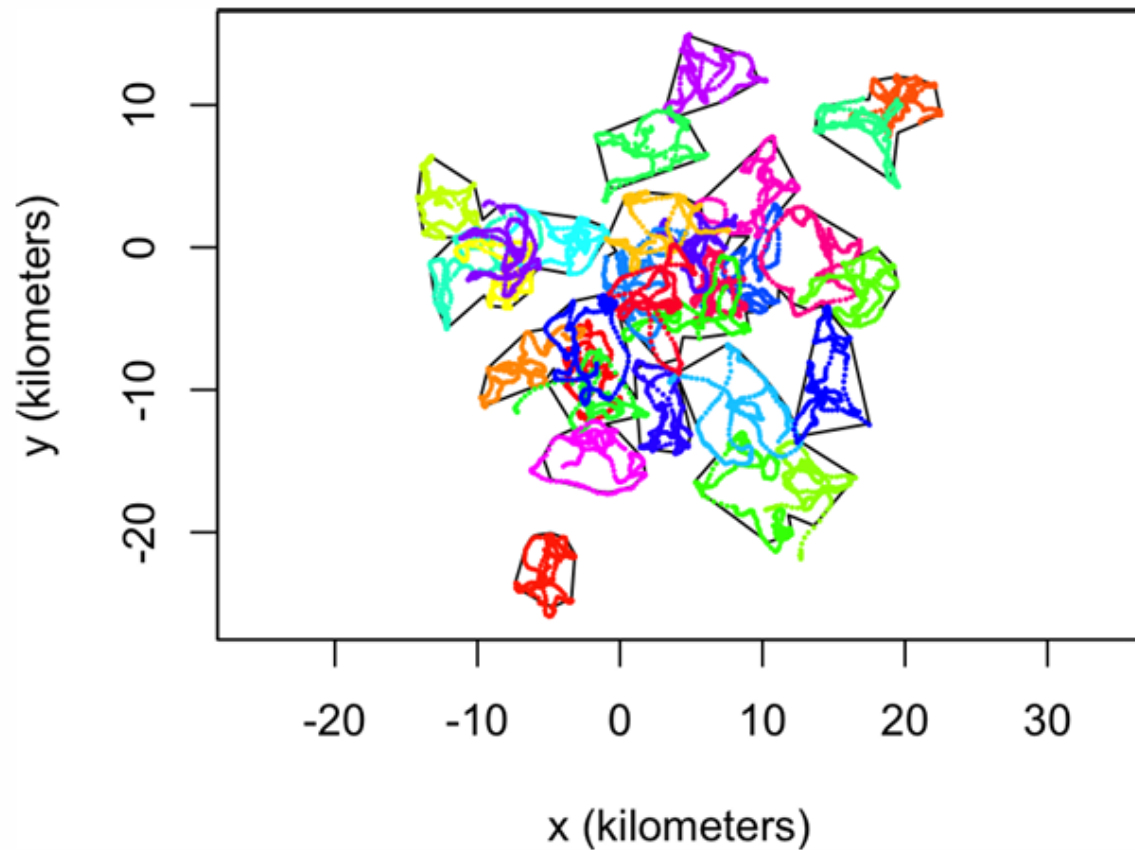
Validation: Simulations

Merged MCP estimate

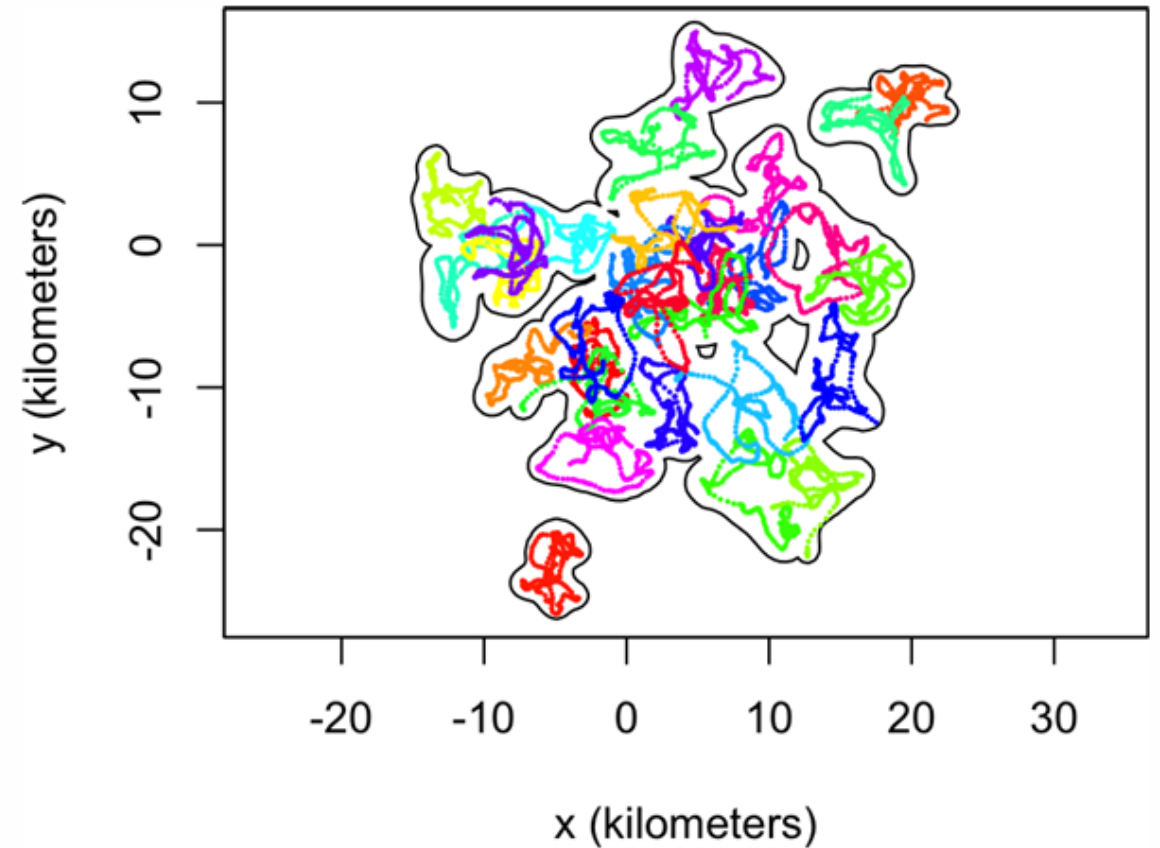


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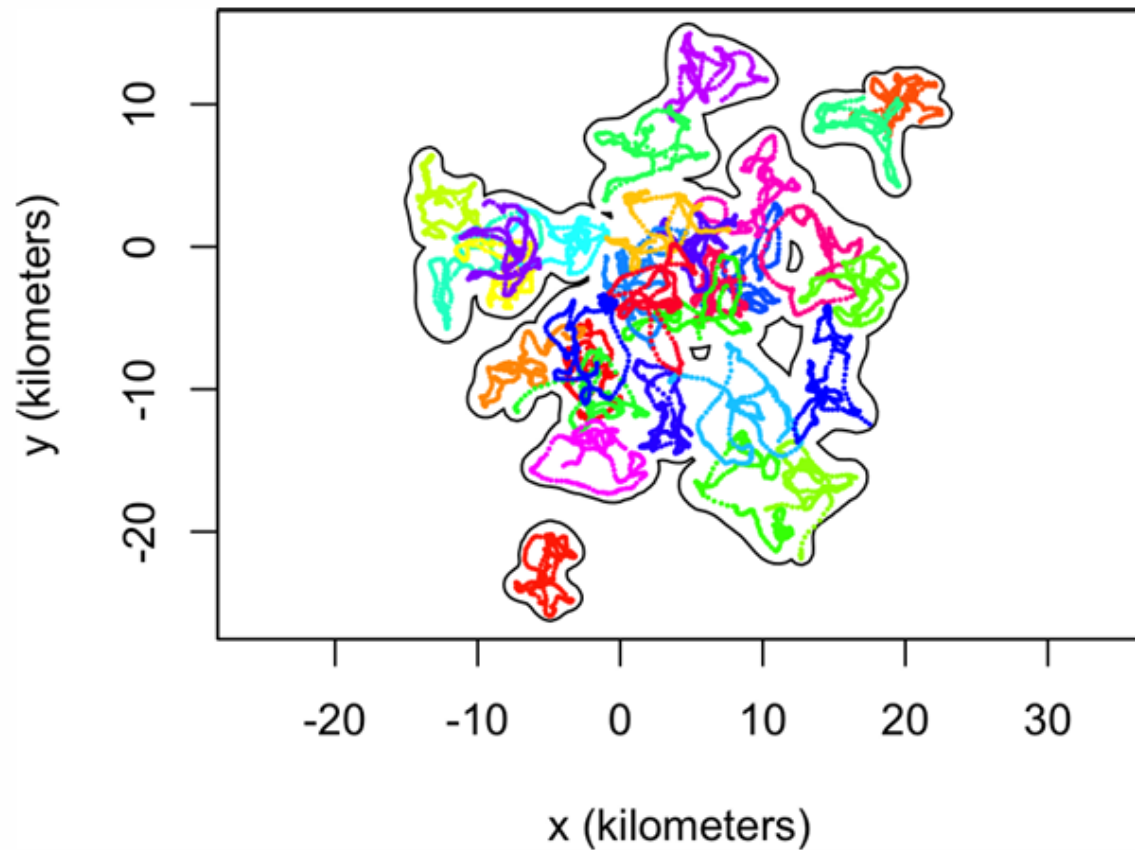


Merged KDE estimate

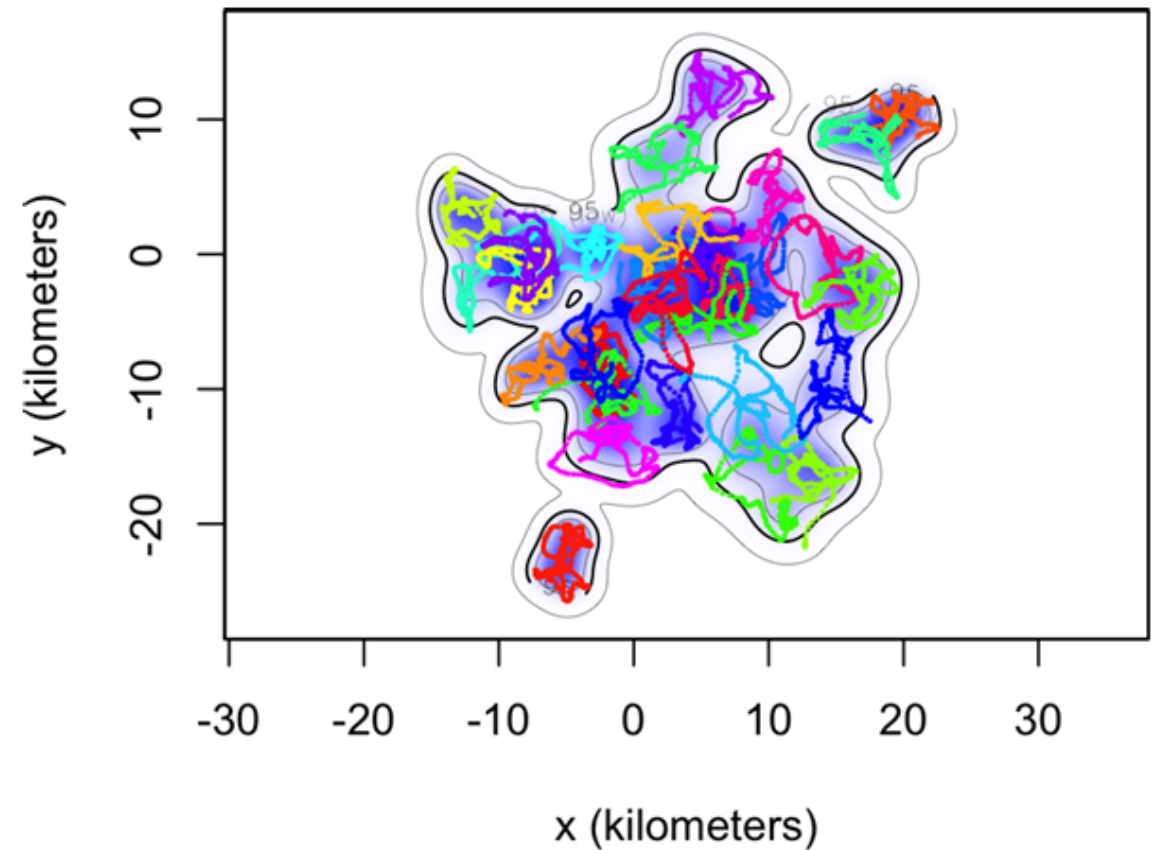


Validation: Simulations

Merged KDE estimate

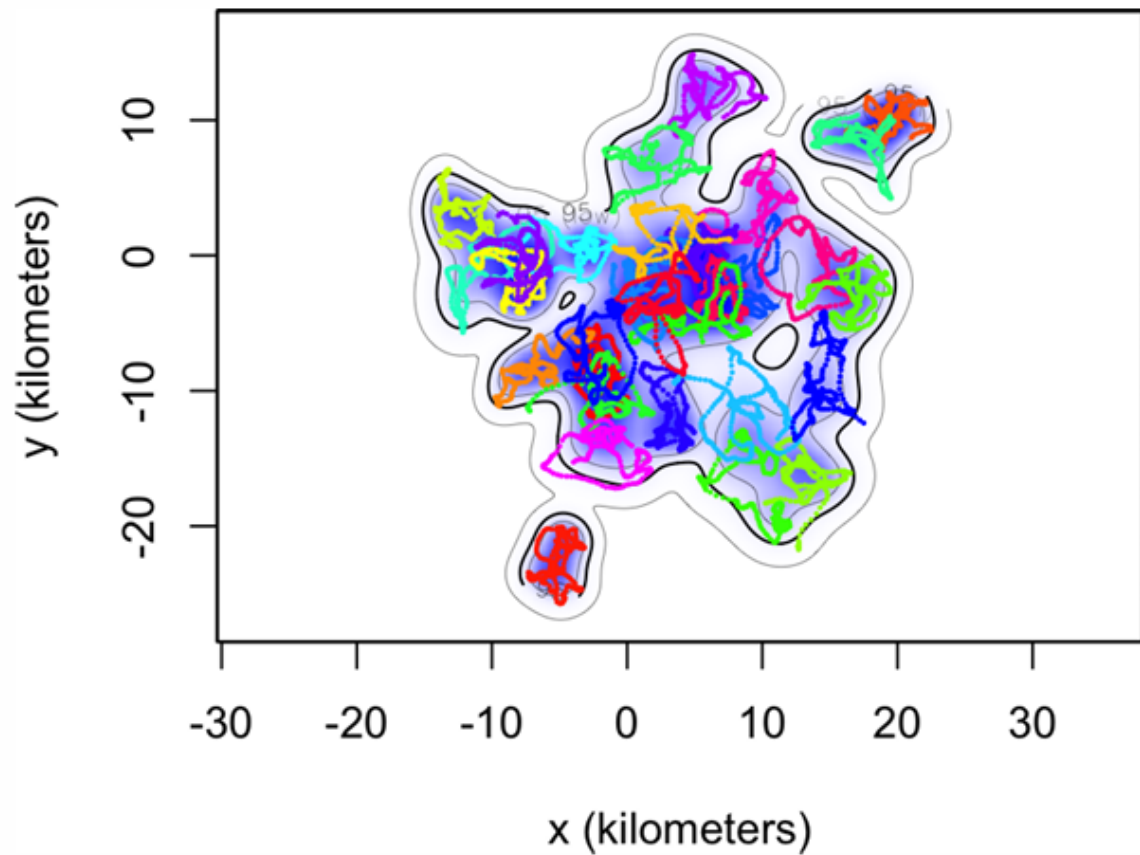


Mean AKDE estimate

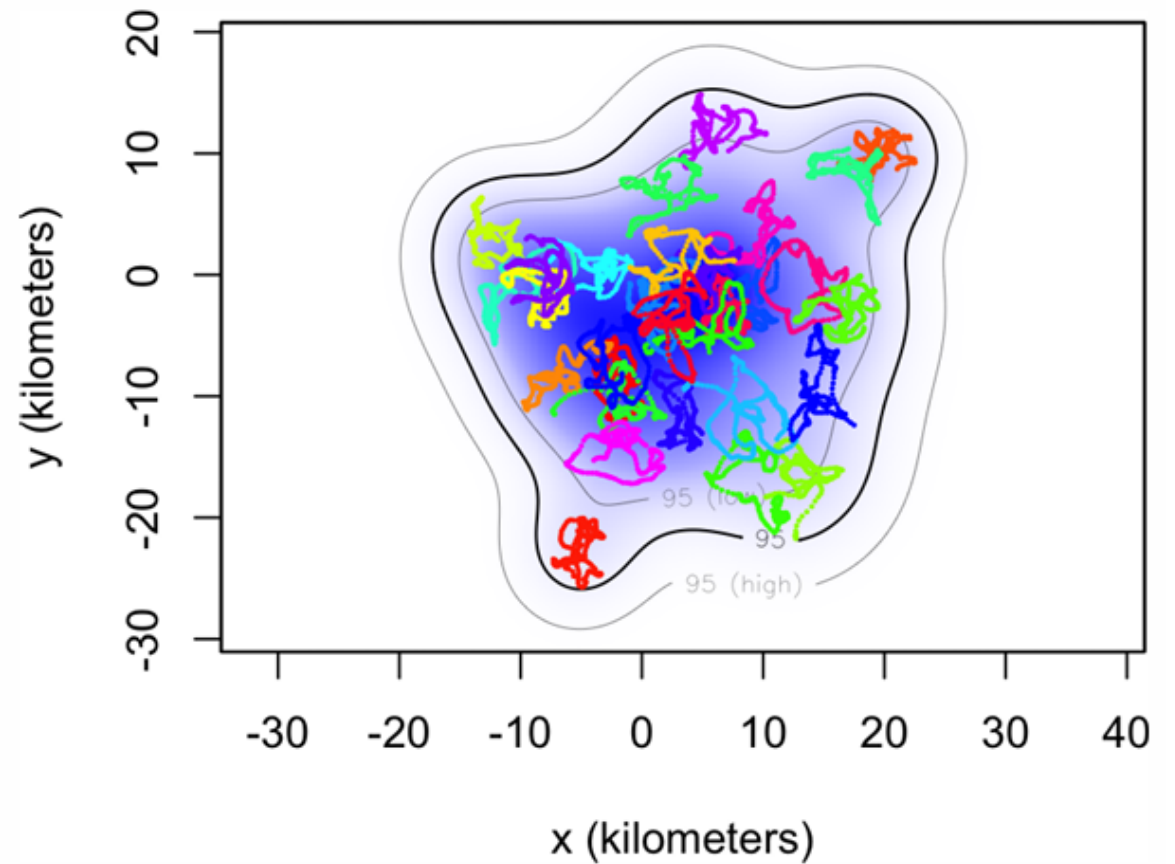


Validation: Simulations

Mean AKDE estimate

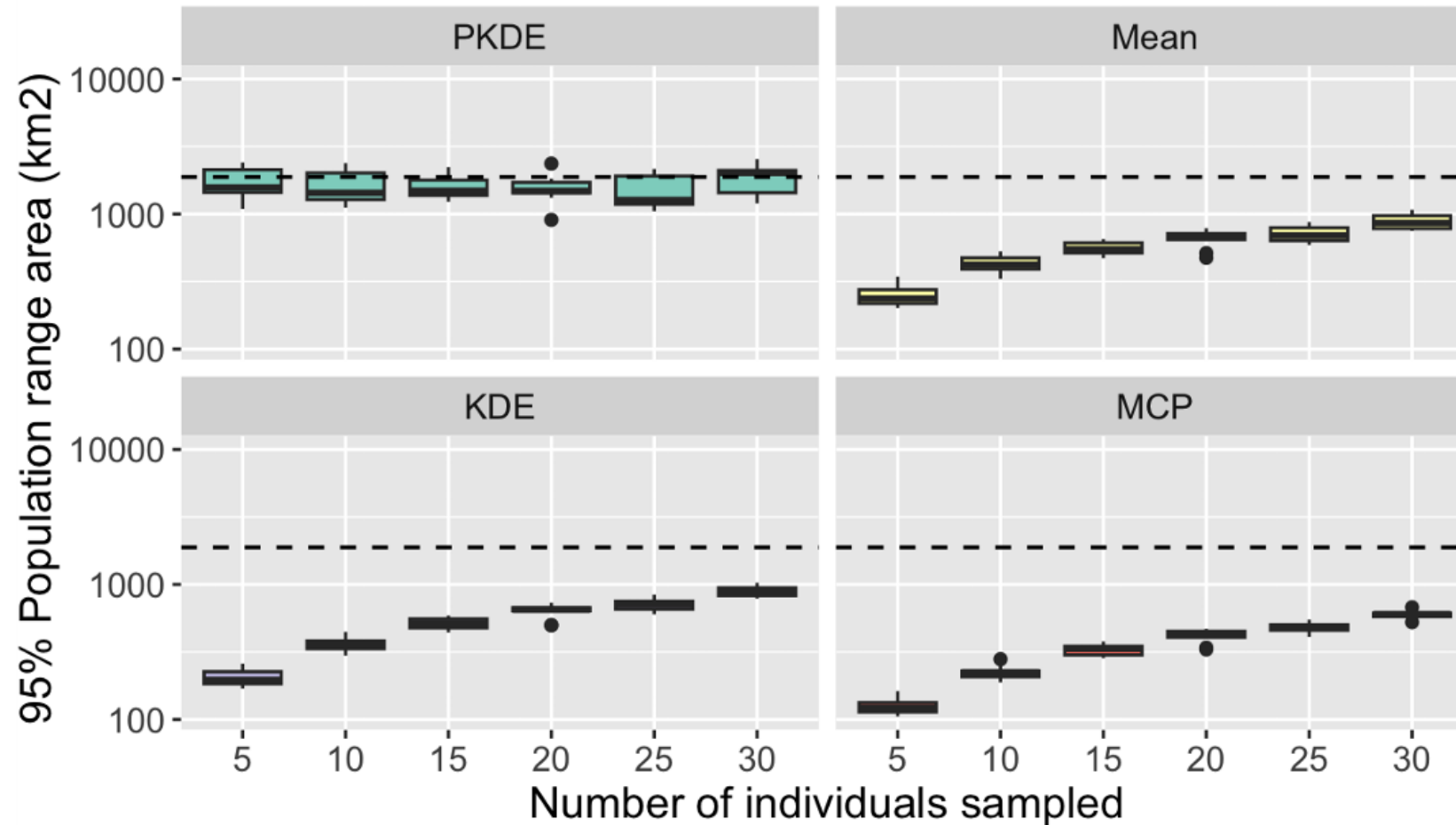


PKDE estimate



Validation: Simulations

Saturation Curve for Simulated Data



Validation: Empirical

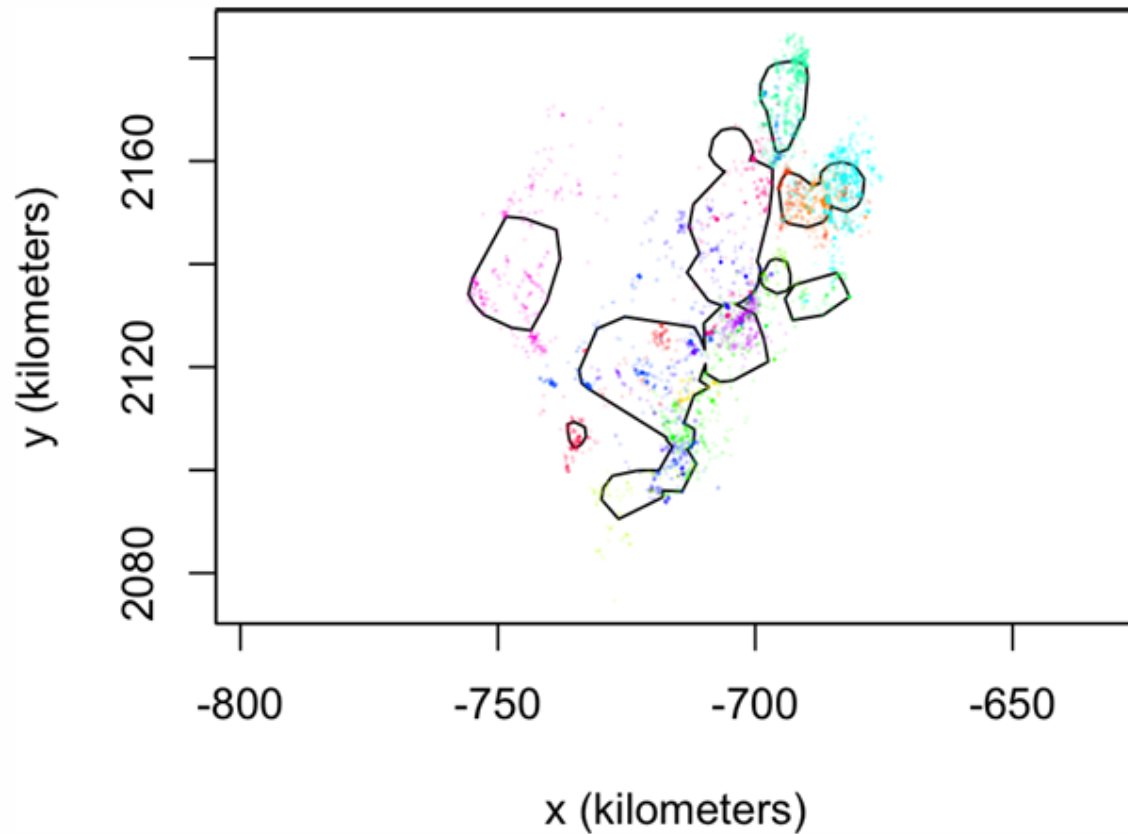
LC

 Paul Graham

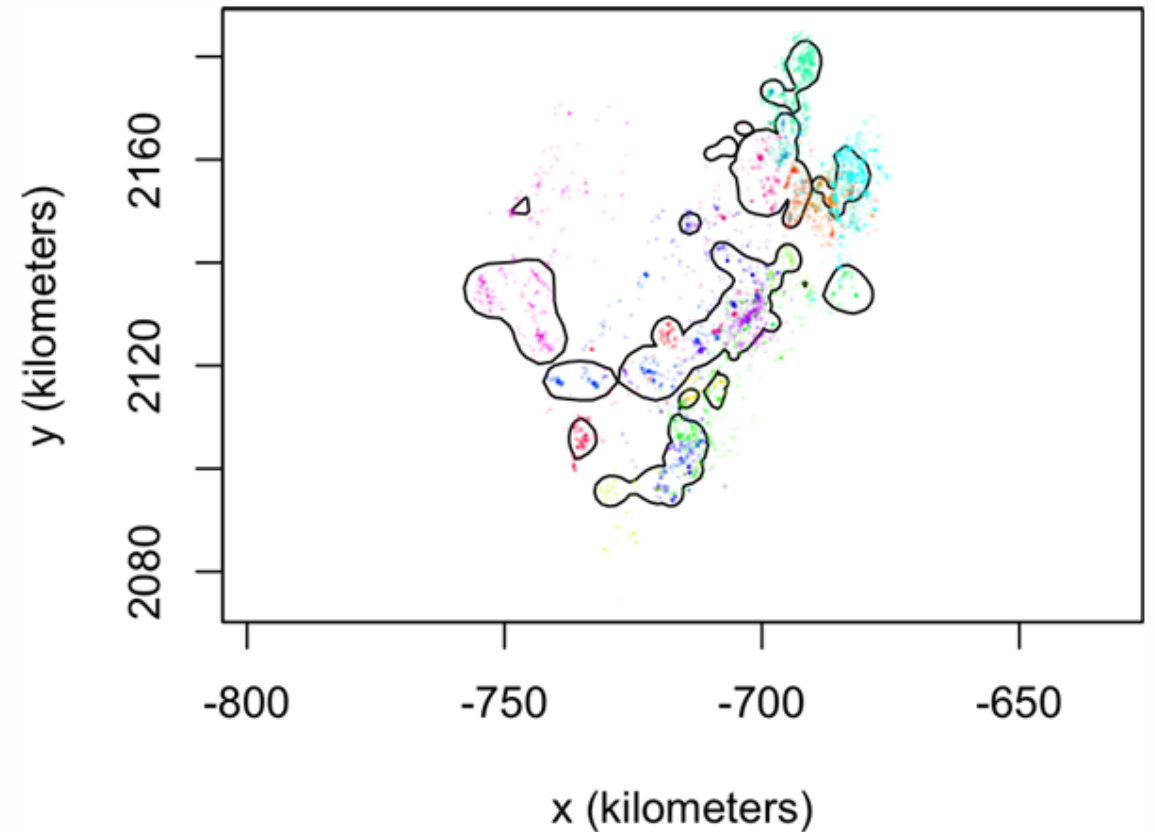
Grizzly Bear
(*Ursus arctos horribilis*)

Validation: Empirical

Merged MCP estimate

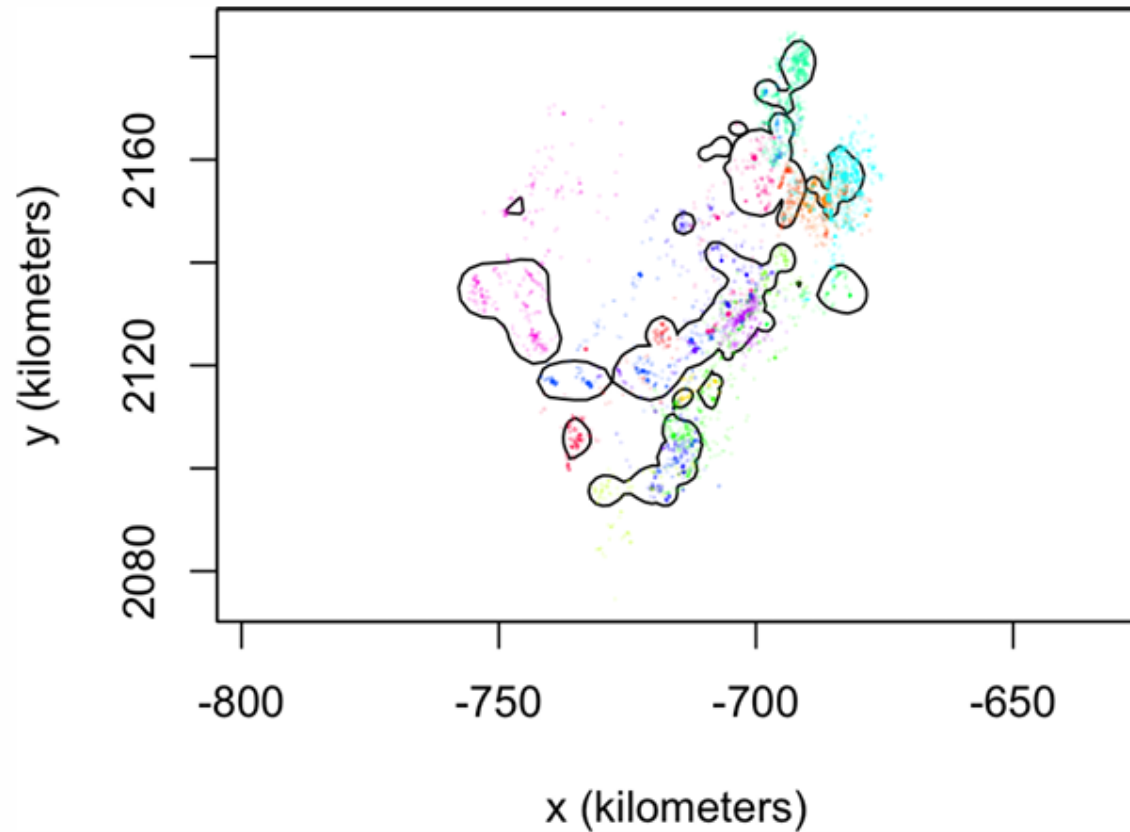


Merged KDE estimate

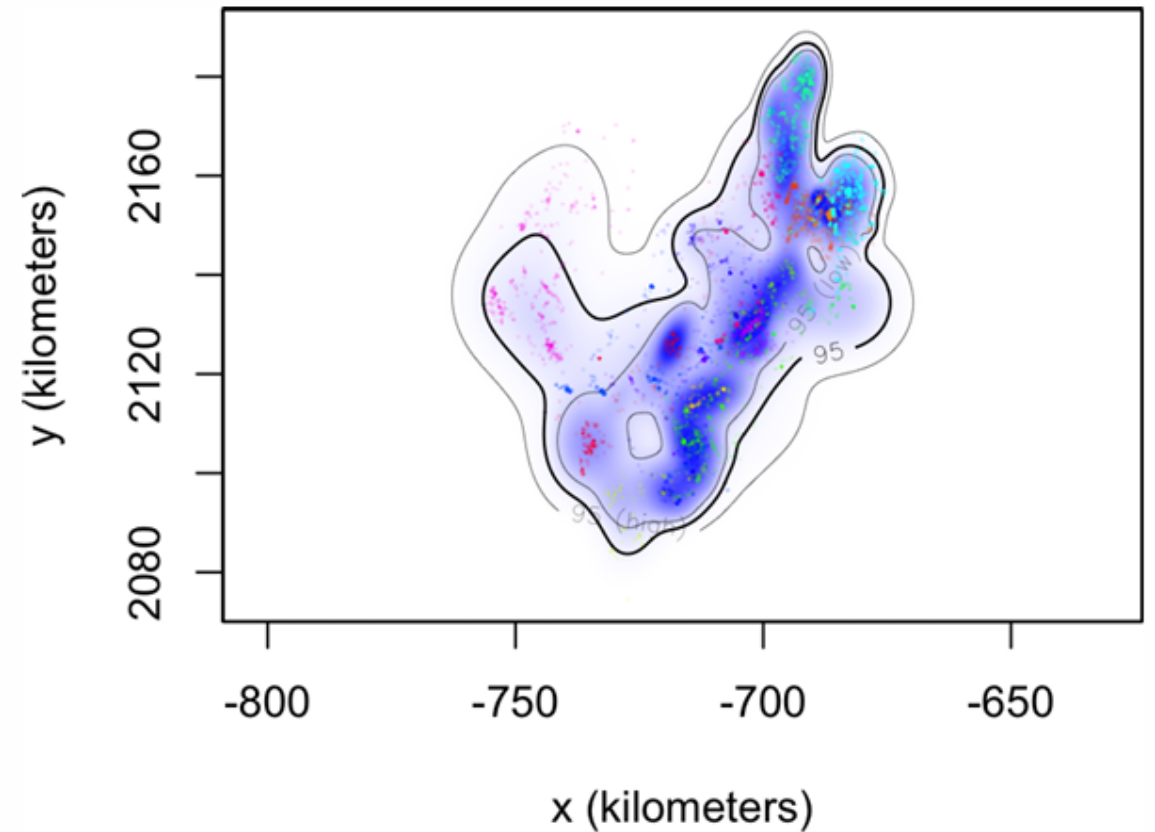


Validation: Empirical

Merged KDE estimate

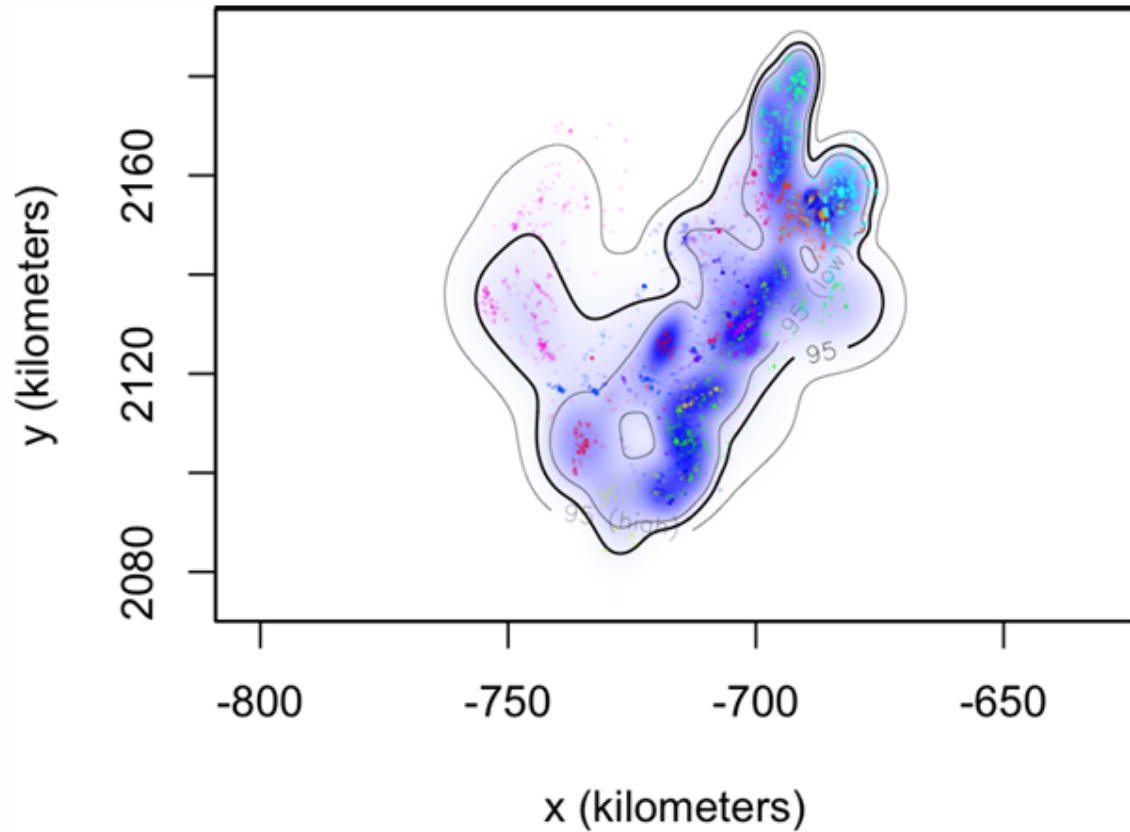


Mean AKDE estimate

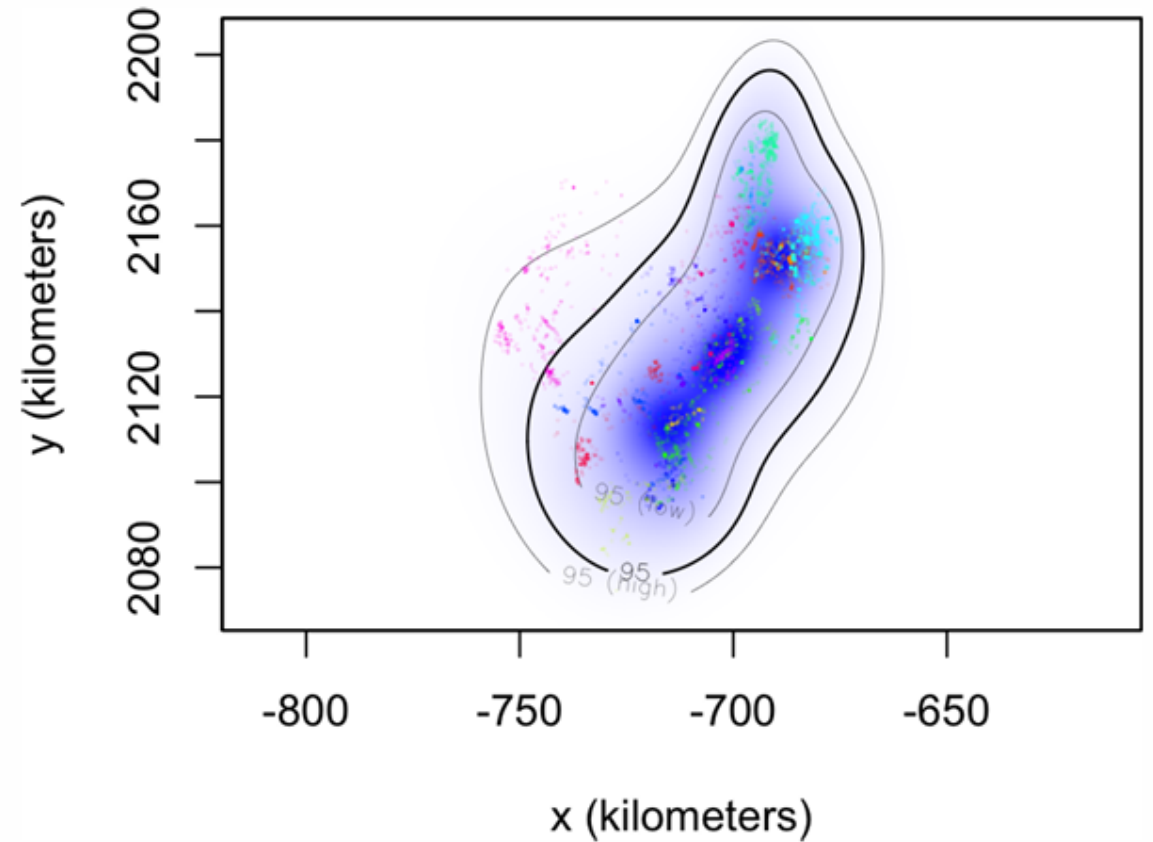


Validation: Empirical

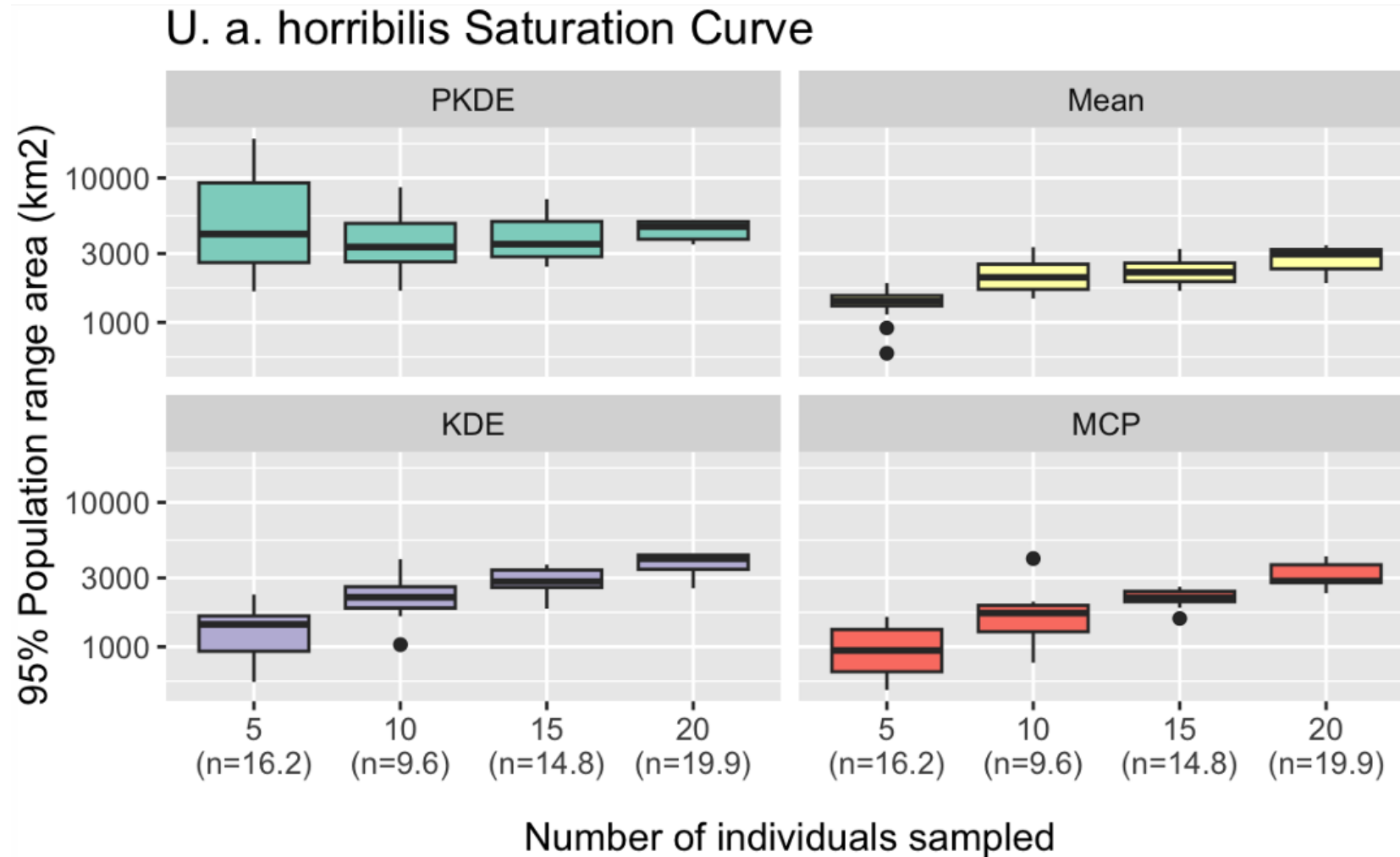
Mean AKDE estimate



PKDE estimate



Validation: Empirical



Overview

- 1) Introduction
- 2) Kernel density estimation
- 3) Validation
- 4) Discussion**

Discussion

Conclusion:

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Future directions:

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Future directions:

- Downweighting poorly sampled individuals: $W_{\text{ind.}} \propto \left(1 + \frac{1}{\text{DOF}[\text{Area}_{\text{ind.}}]}\right)^{-1}$
- Downweighting non-representative individuals: $W_{\text{ind.}} \propto \langle p_{\text{ind.}} | p_{\text{pop.}} \rangle$



ctmm_akde.R

Thank you

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